Digital Image Processing:

2 Digital Imaging Fundamentals

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Contents

This lecture will cover:

1. The human visual system
2. Light and the electromagnetic spectrum
3. Image sensing and acquisition
4. Image sampling, quantization and representation
5. Image resolution

1 Human Visual System

The best vision model we have!

✓ Basic understanding of human visual perception is important; Aim to know:

a. The mechanics and parameters related to how images are formed in the eye;

b. The physical limitations of human vision in terms of factors that also are used in digital image processing;

c. The factors how human and electronic imaging compare in terms of resolution and ability to adapt to changes in illumination.

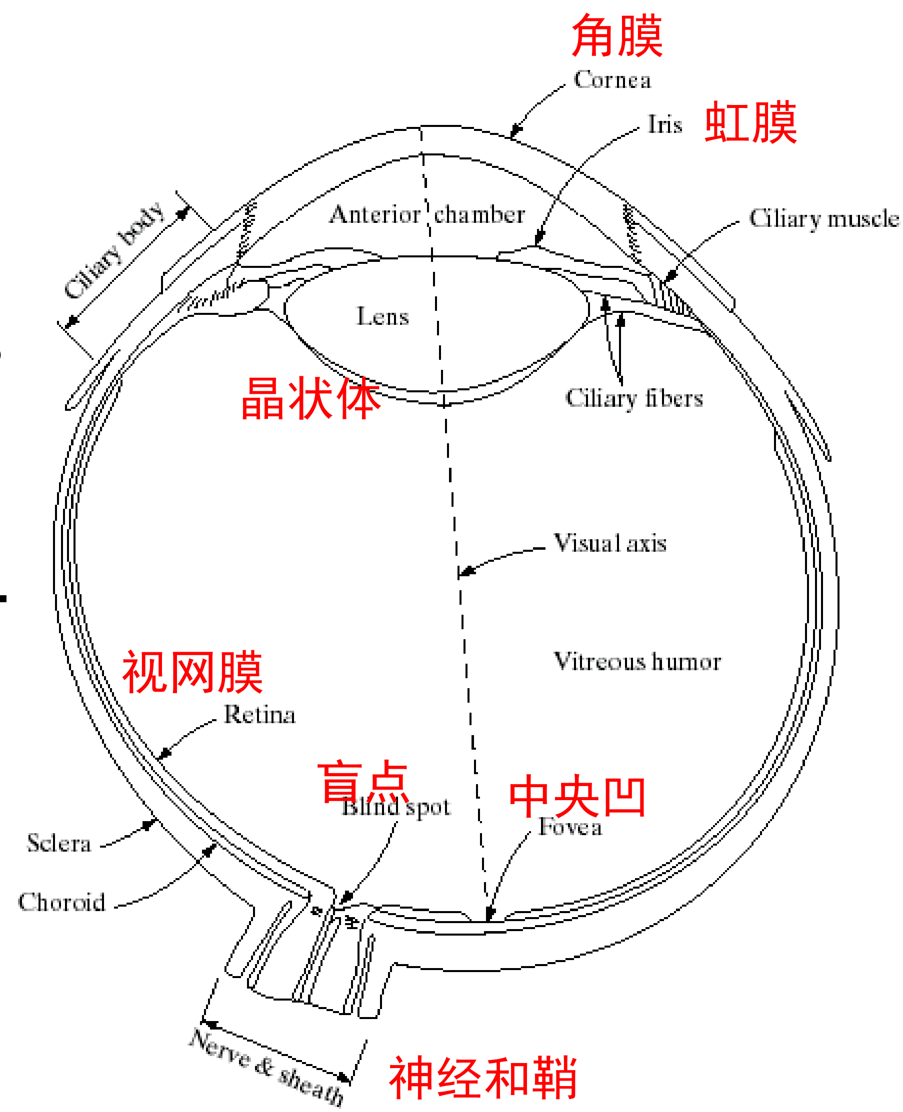
1.1 Structure Of The Human Eye

Lens: focuses light from objects onto the retina(视网膜).

- can be contracted
- shape of lens is varied to focus on objects at different distances
- zoom (to a plane)
- IR and UV light are absorbed by proteins in the lens structure.

Pupil:

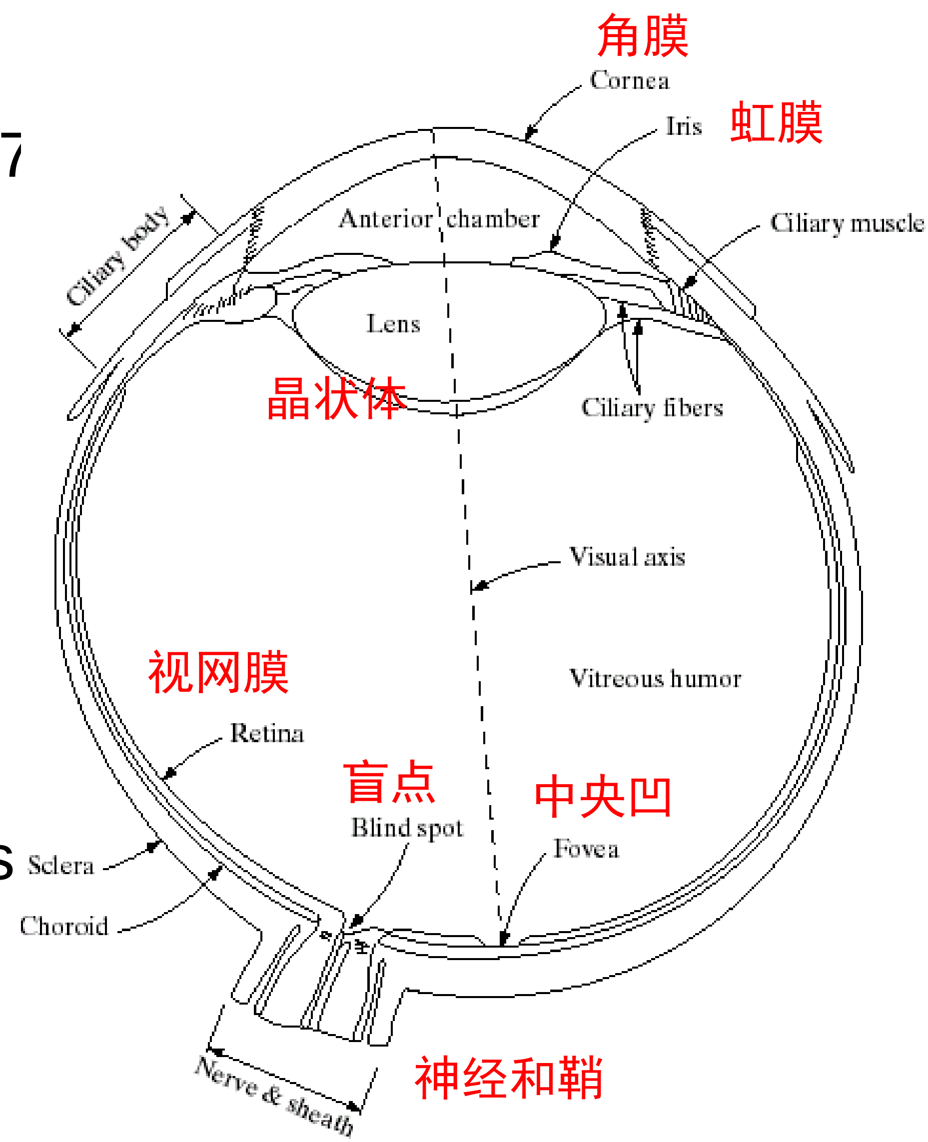
- Open varies from 2-8mm
- Regulates the amount of light reaching the retina.
- Same as aperture of a camera.



1.1 Structure Of The Human Eye

◆Retina is covered with light receptors called *cones* ***锥状体*** (6-7 million) and *rods* ***杆状体*** (75-150 million).

- Cones are concentrated around the fovea, highly sensitive to color.
- Rods are more spread out and are sensitive to low levels of illumination.
- Under moonlight, colorless



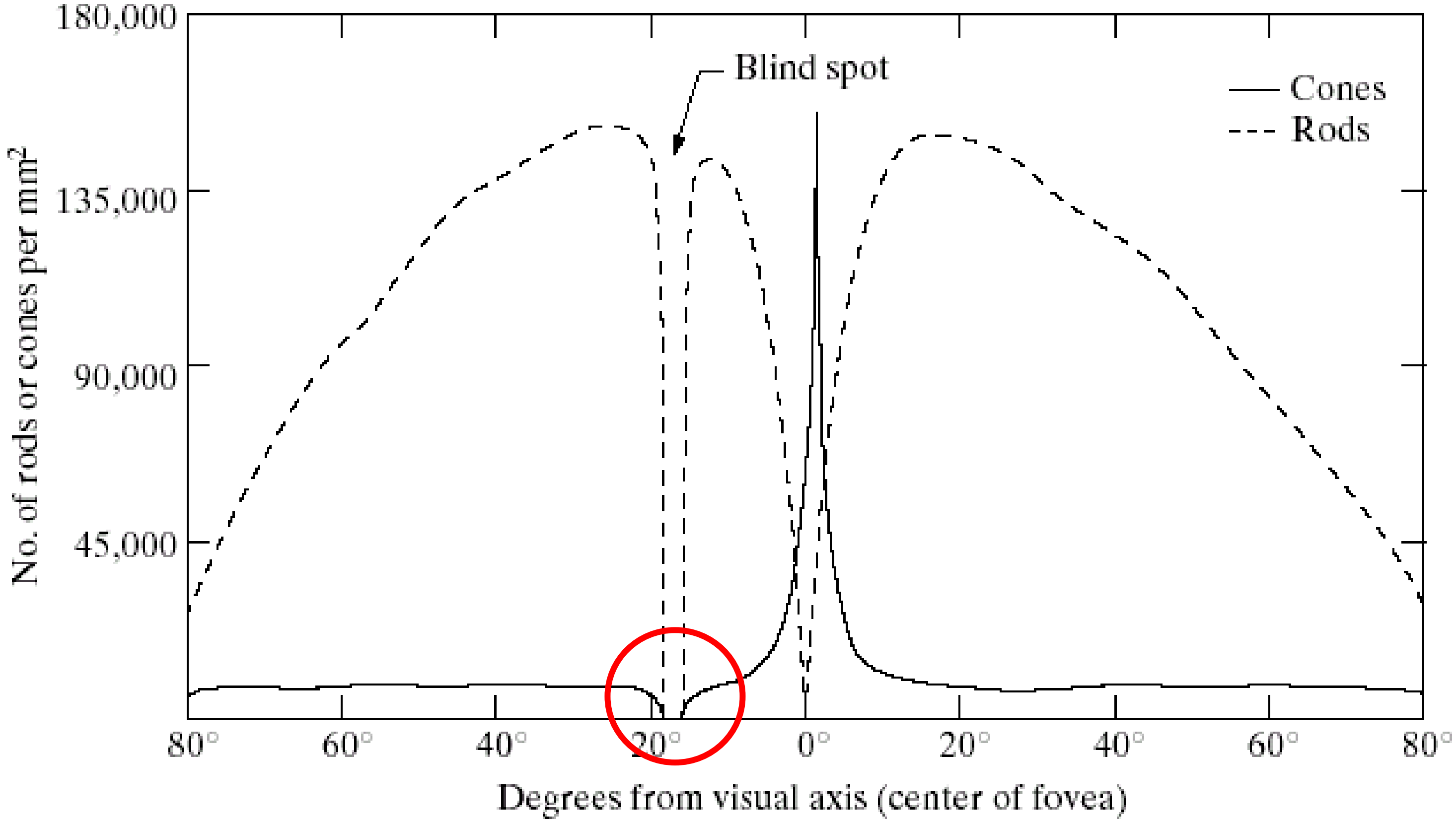


FIGURE 2.2 Distribution of rods and cones in the retina.

Woman or Man have more cones?

Can see more color!

Blind-Spot Experiment

➢ Prepare a drawing similar to that below on a piece of paper (the dot and cross are about 15cm apart);

➢ Close your right eye and focus on the cross with your left eye. Hold the image about 50cm away from your face and move it slowly towards you.

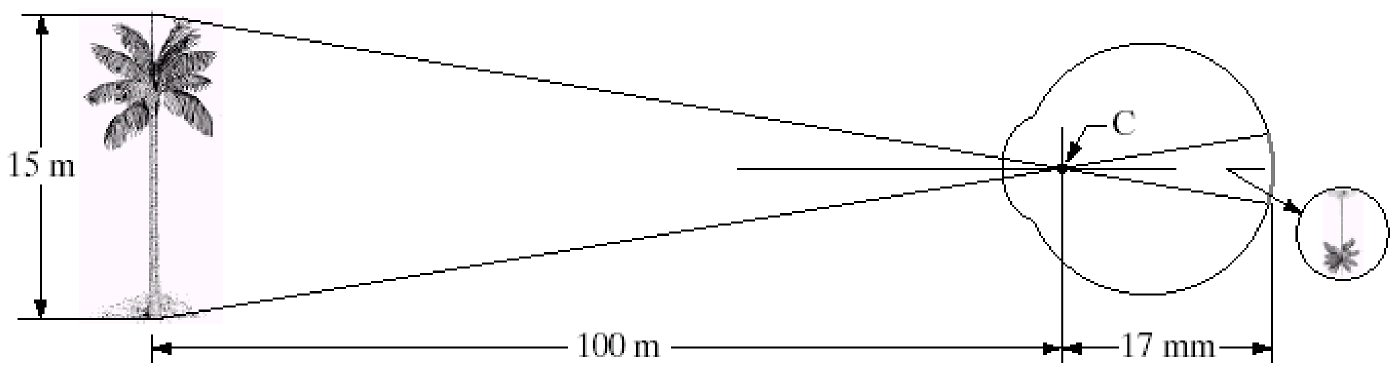
➢ What's happened? The dot should disappear!

➢The eye's retina receives and reacts to incoming light and sends signals to the brain, allowing you to see. One part of the retina, doesn't give you visual information—this is your eye's "**blind spot**."

1.2 Image Formation In The Eye

Far and near: Muscles within the eye can be used to change the shape of the lens allowing us focus. (*Flattening or thickening the lens*)

How: An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain.



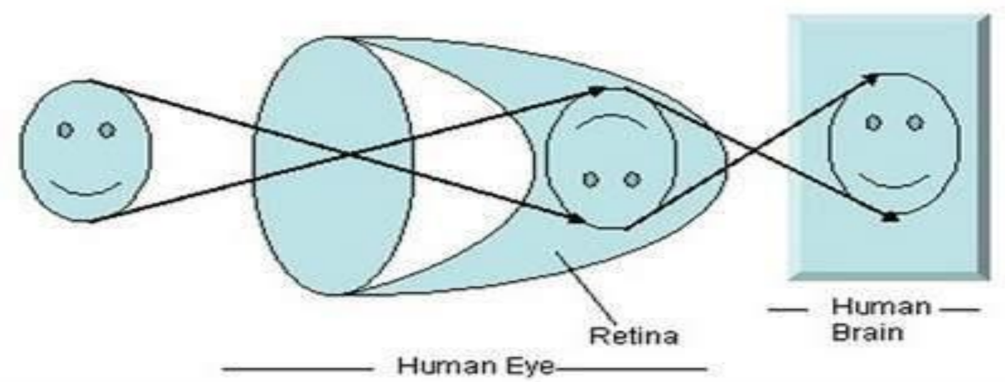
Focal length: 14-17mm   Length of tree image≅2.55mm

For distant objects (>3m), lens exhibits the least refractive power (flattened)

For nearby objects (<1m), lens is most strongly refractive (curved)

Q: which distance (distant or nearby) happens when the eye is relaxed, why?

How human eye works?



- When sun light falls on the object (in this case the object is a face), it is reflected.
- Different rays form different angle when they are passed through the lens and an invert image of the object has been formed on the back wall.
- The last portion of the figure denotes that the object has been interpreted by the brain and re-inverted.

1.3 Brightness Adaptation & Discrimination

Image Formation Model $f(x,y)=i(x,y)r(x,y)$

$0<f(x,y)<\infty$ luminance – proportional to energy radiated by a physical source

$0<i(x,y)<\infty$ Illumination

$0<r(x,y)<1$ reflectance

f(x,y) r(x,y) i(x,y)

1.3 Brightness Adaptation & Discrimination

Lightness Perception: ***Objective*** Quantities 客观

- *Luminance* is the amount of visible light that comes to the eye from a surface.
- *Illuminance* is the amount of light incident on a surface.
- *Reflectance* (also called *albedo*) is the proportion of incident light that is reflected from a surface.
  - varies from 0% to 100% where 0% is ideal black and 100% is ideal white. In practice, typical black paint is about 5% and typical white paint about 85%.

1.3 Brightness Adaptation & Discrimination

Lightness Perception: ***Subjective*** Quantities 主观

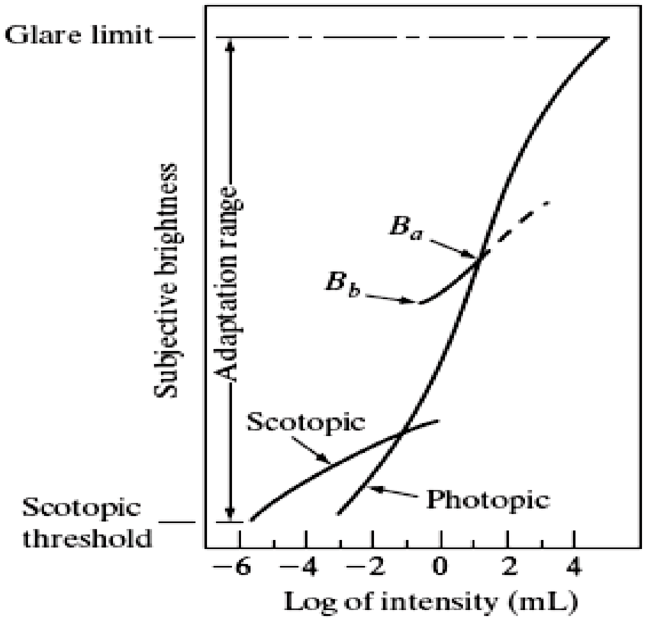
- ***Lightness*** is the perceived reflectance of a surface. It represents the visual system's attempt to extract reflectance based on the luminance in the scene.

- ***Brightness*** is the perceived intensity of light coming from the image itself, rather than any property of the portrayed scene. Brightness is sometimes defined as perceived luminance.

1.3 Brightness Adaptation & Discrimination

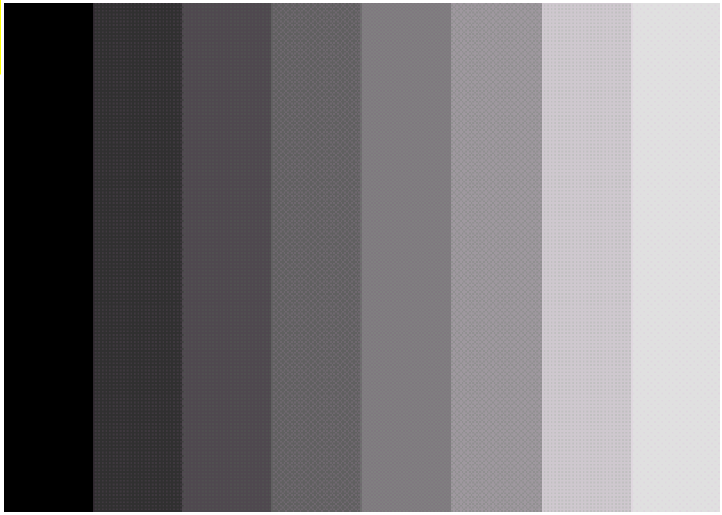
- The human visual system can perceive approximately 10^{10} different light intensity levels.
- Subjective brightness of human is a log function of the light intensity.
- At any one time we can only **discriminate between a much smaller number** – *brightness adaptation*.

- **Brightness discrimination**
  - poor at low levels of illumination,
  - better with increasing illumination



Brightness Adaptation & Discrimination (cont…)

Human *perceived brightness* is not a simple function of intensity!-Some interesting phenomenon:



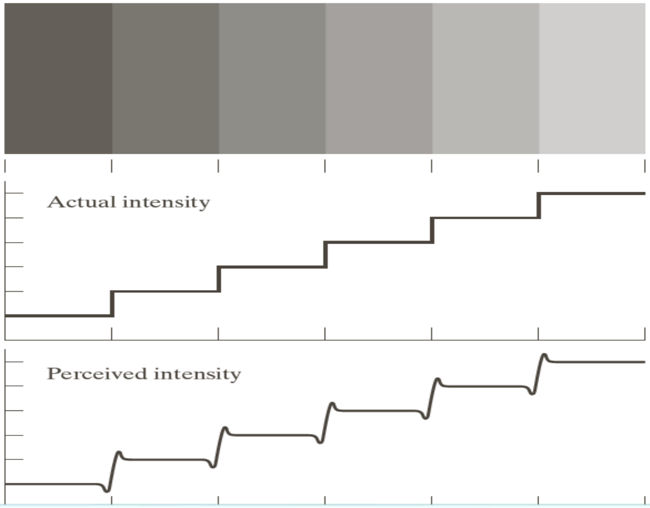
An example of Mach bands

(1) Mach bands

- Stripes appear darker near a more intense stripe (and vice versa) around the boundary of different intensity.
- Caused by inhibitory neural connections.

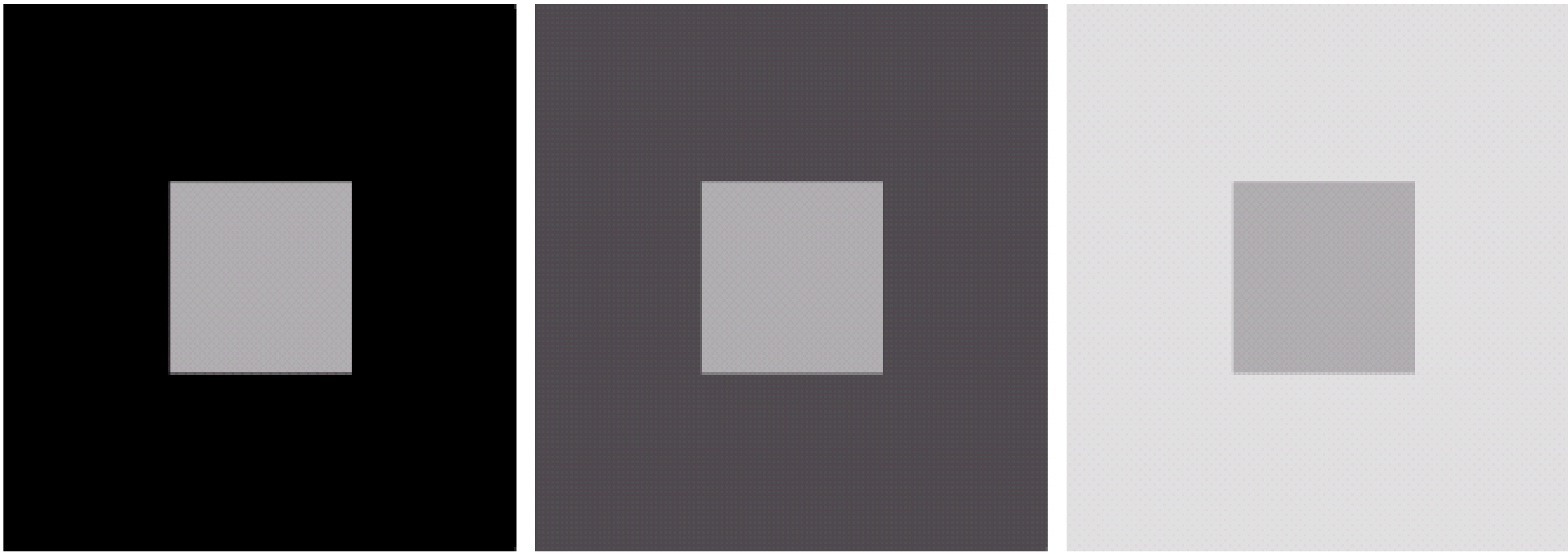
(2) simultaneous contrast

The *perceived intensity* of a region is related to the light intensities of the regions surrounding it.



Gonzalez & Woods, Digital Image Processing (2010)

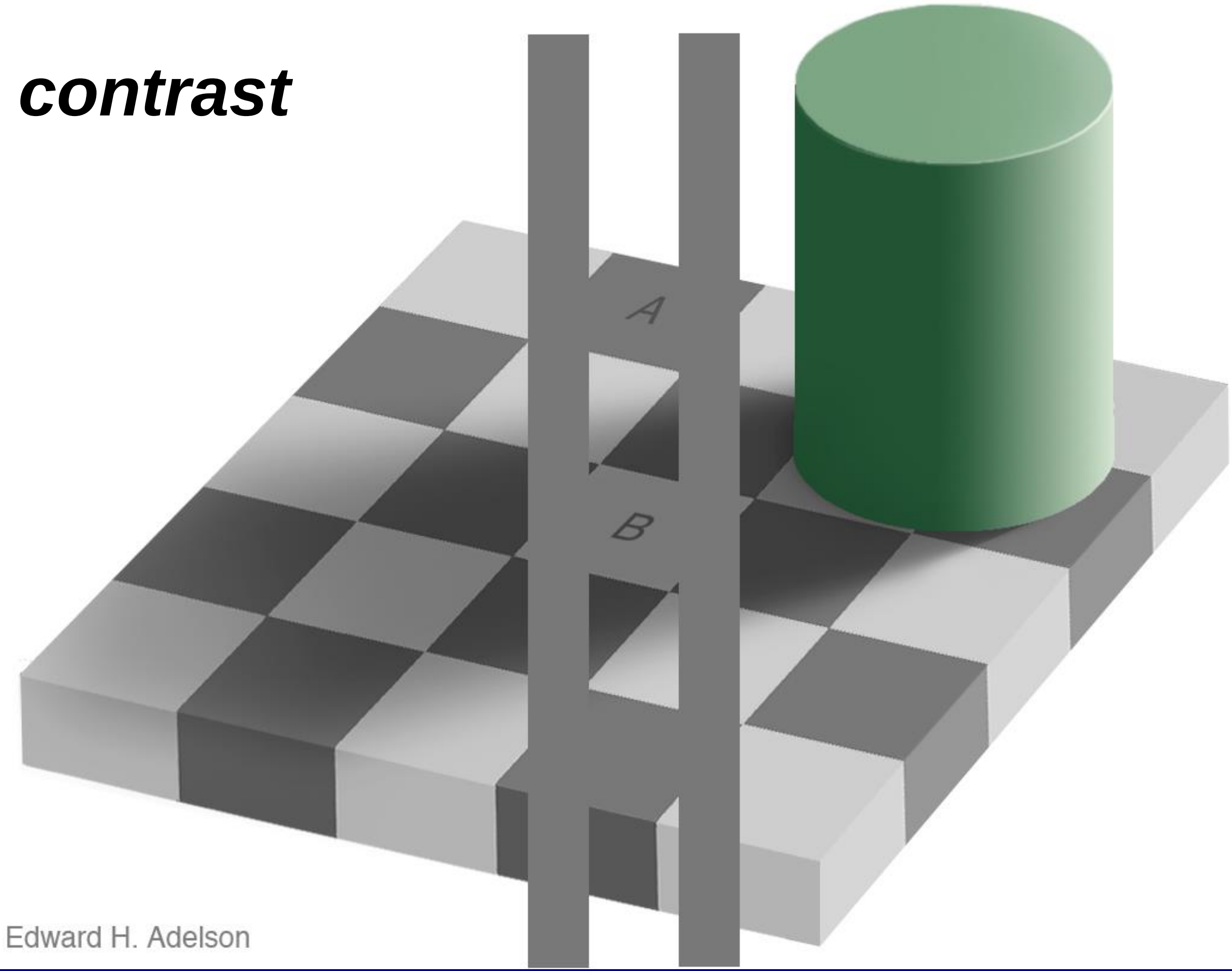
Brightness Adaptation & Discrimination (cont…)



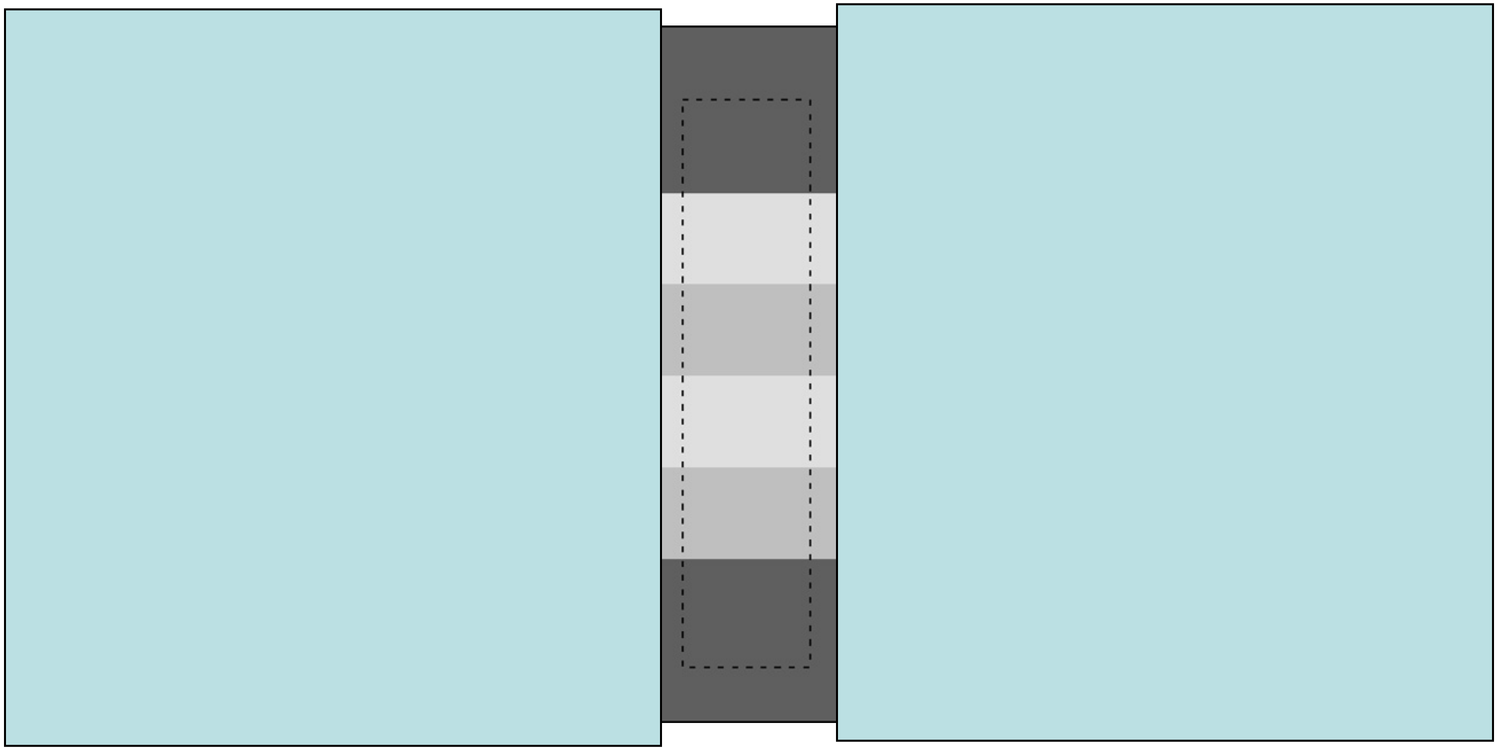
An example of ***simultaneous contrast***

Brightness Adaptation & Discrimination (cont…)

simultaneous contrast

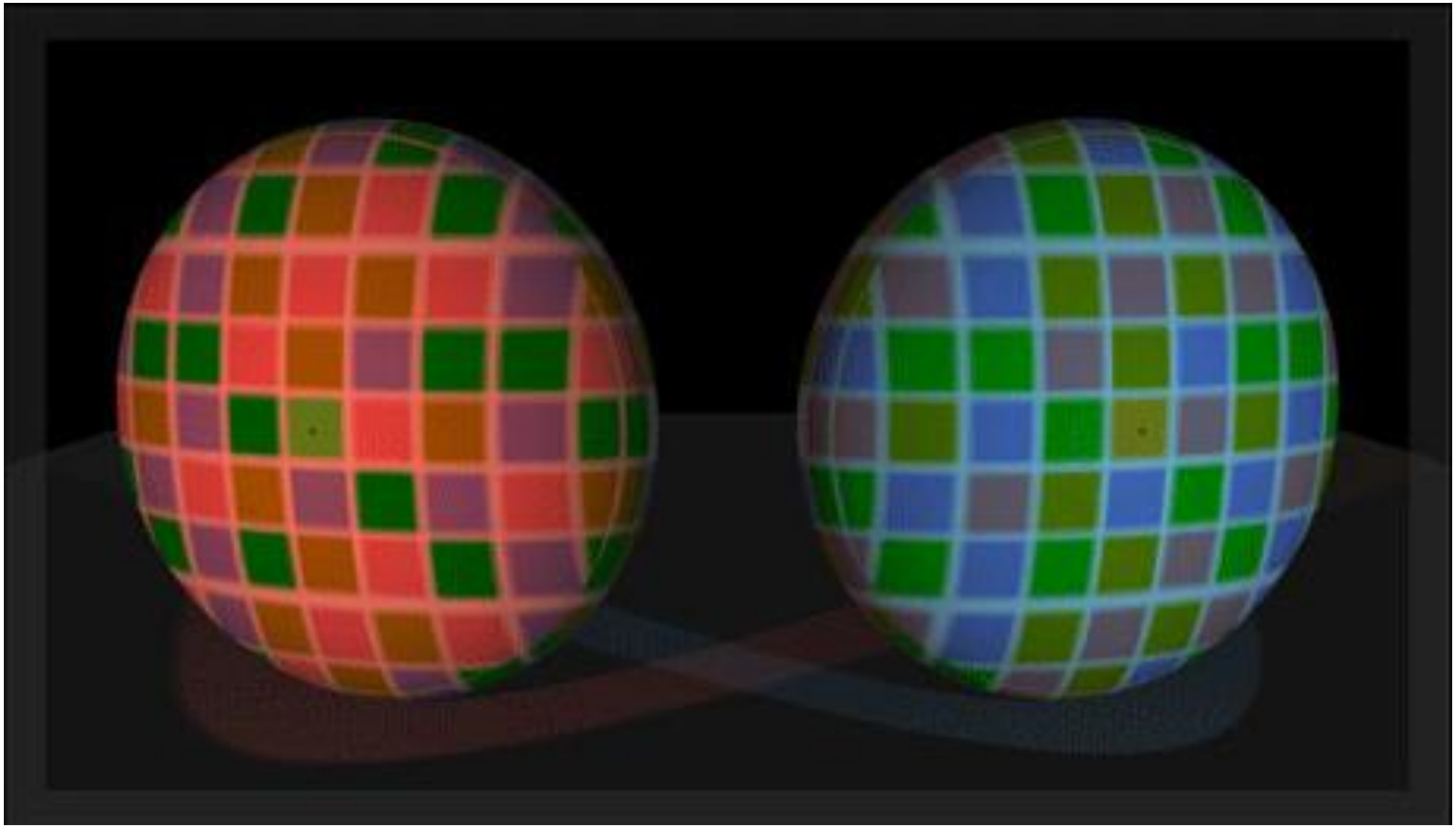


simultaneous contrast



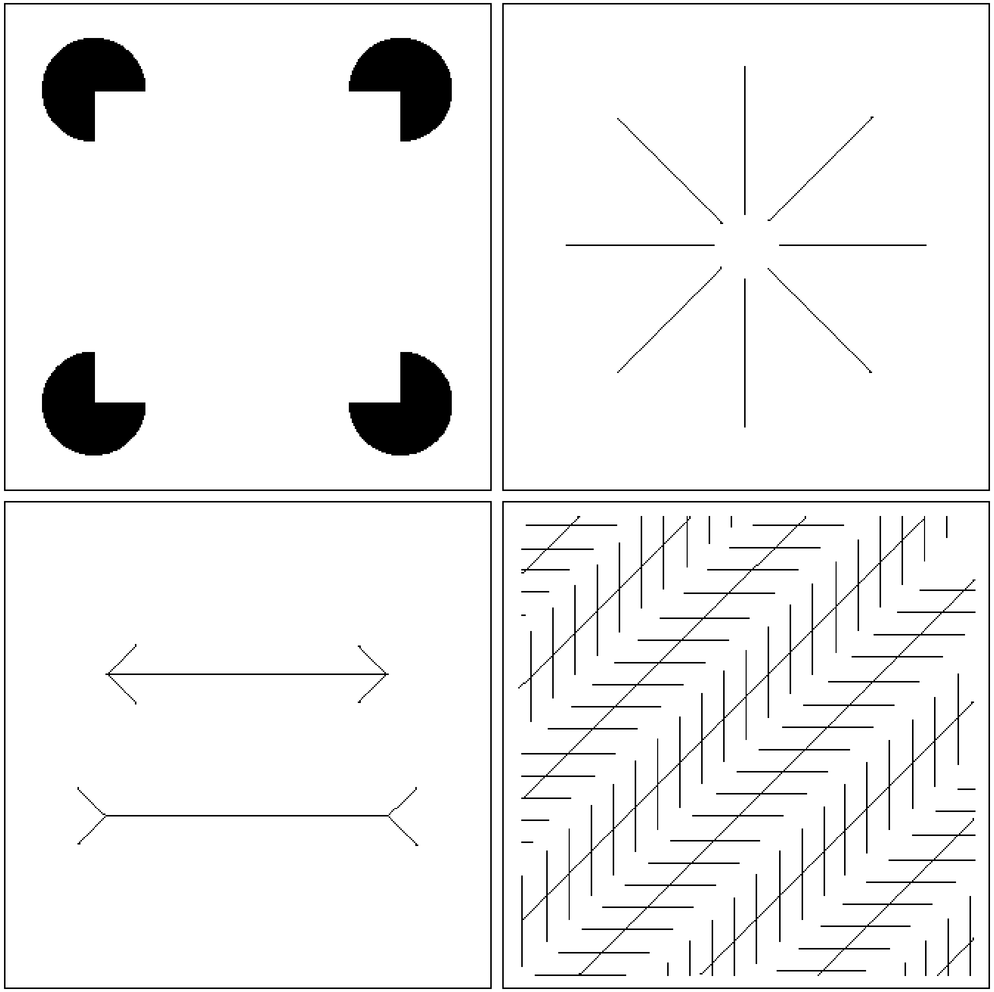
If we cover the right side of the figure and view the left side, it appears that the stripes are due to paint (reflectance). If we cover the left side and view the right, it appears that the stripes are due to different lighting on the stair steps (illumination).

simultaneous contrast

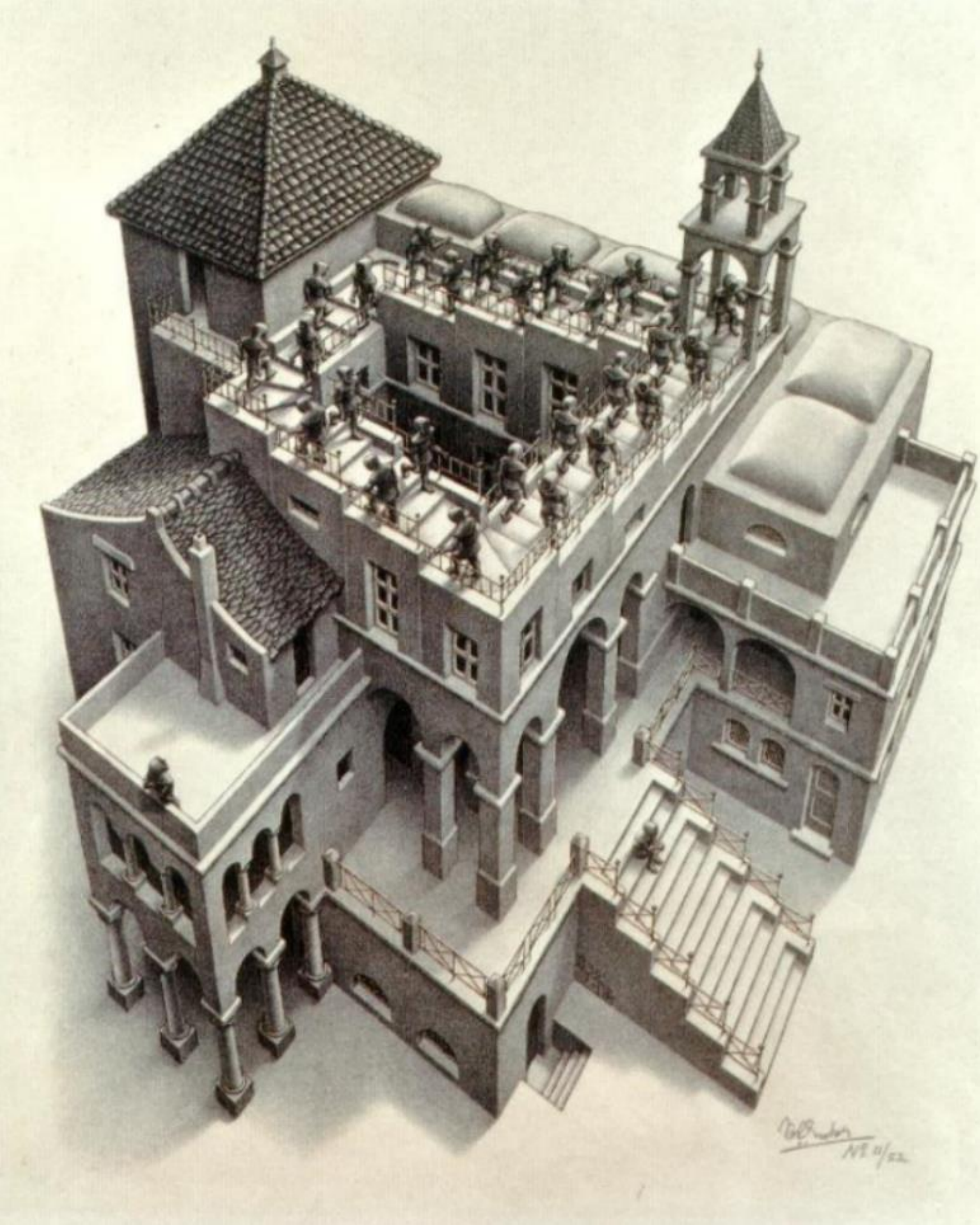


(3) Optical Illusions

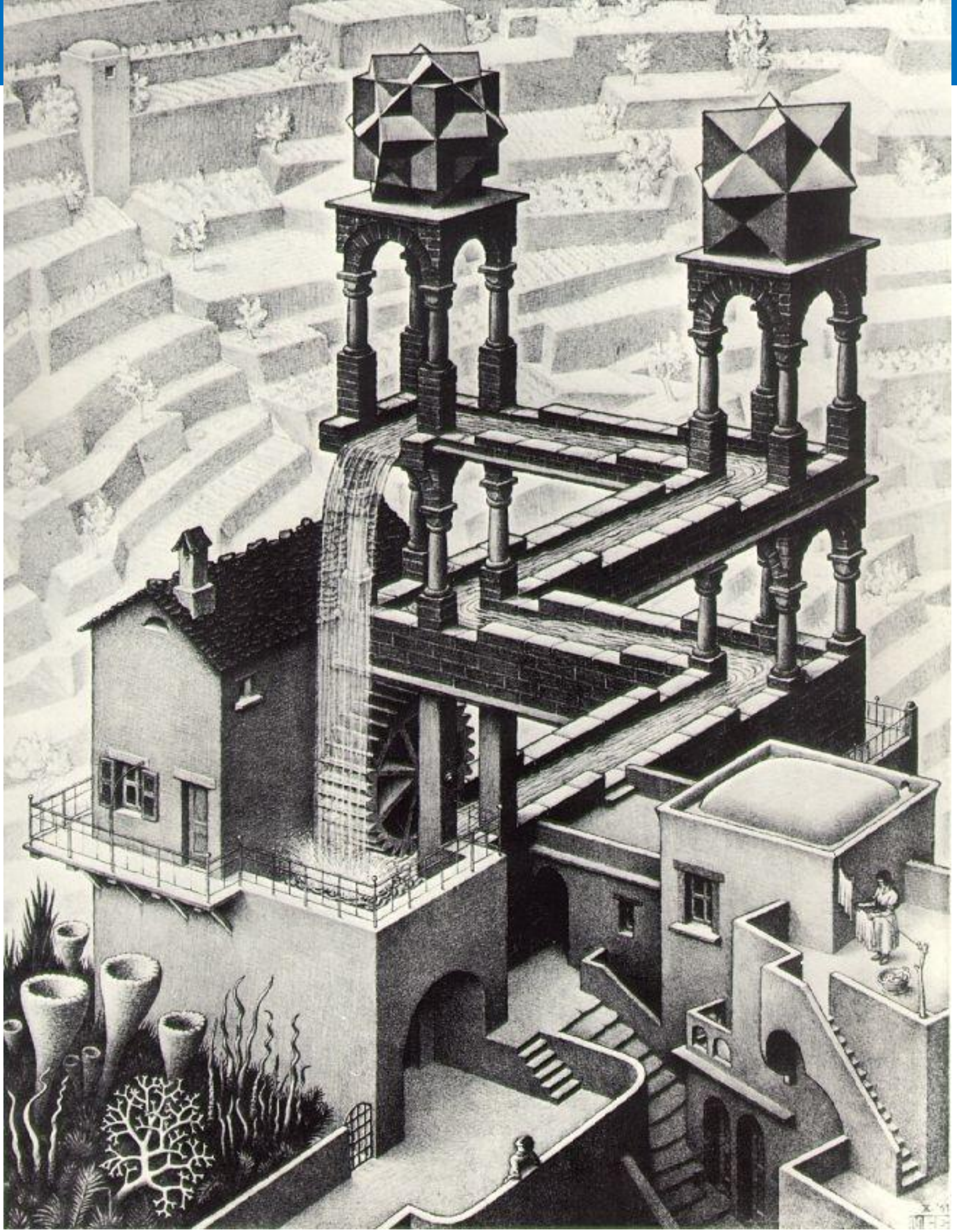
- Our visual systems play lots of interesting tricks on us；
- The eye fills in non-existing information;
- Wrongly perceives geometrical properties of objects.



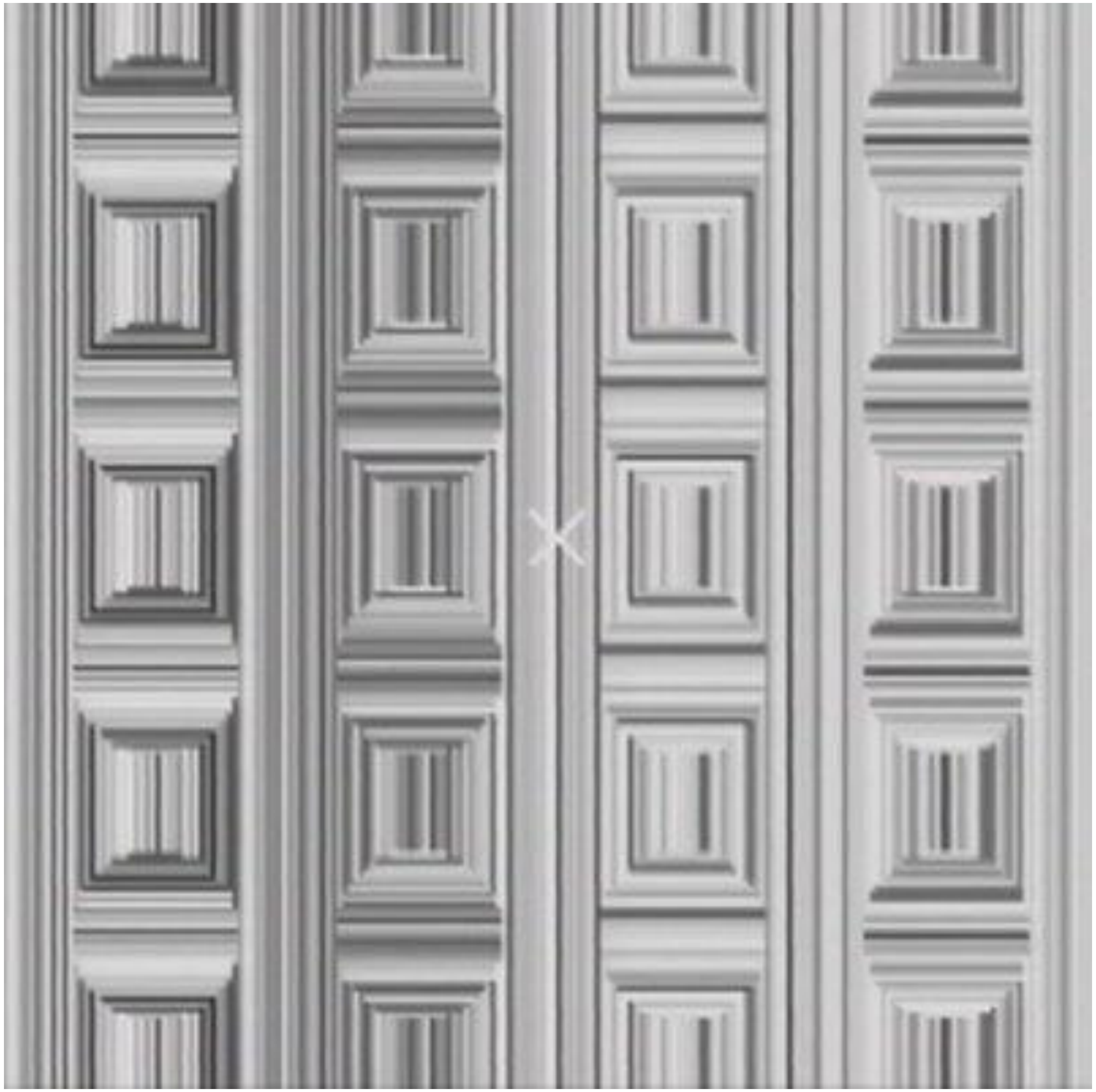
Optical Illusions (cont…)



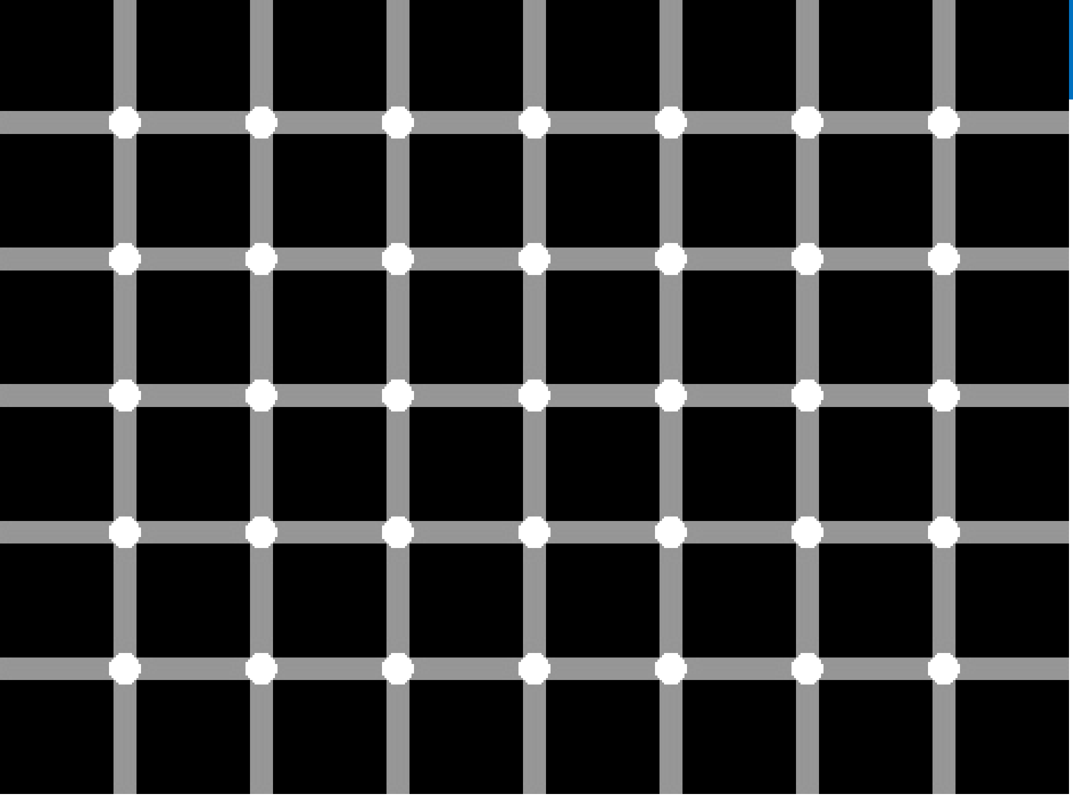
Optical Illusions (cont…)

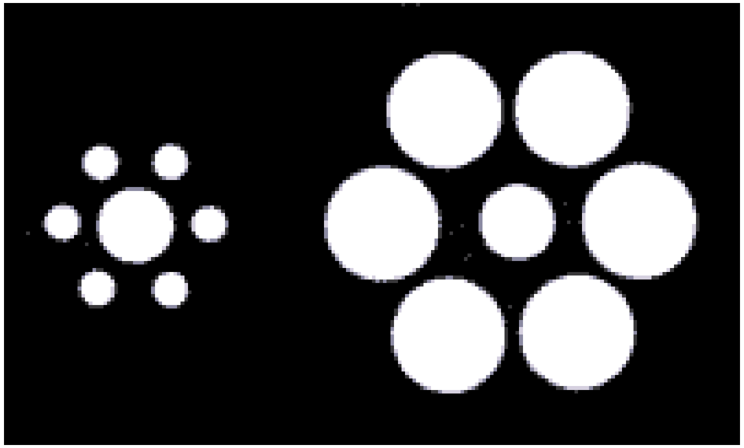


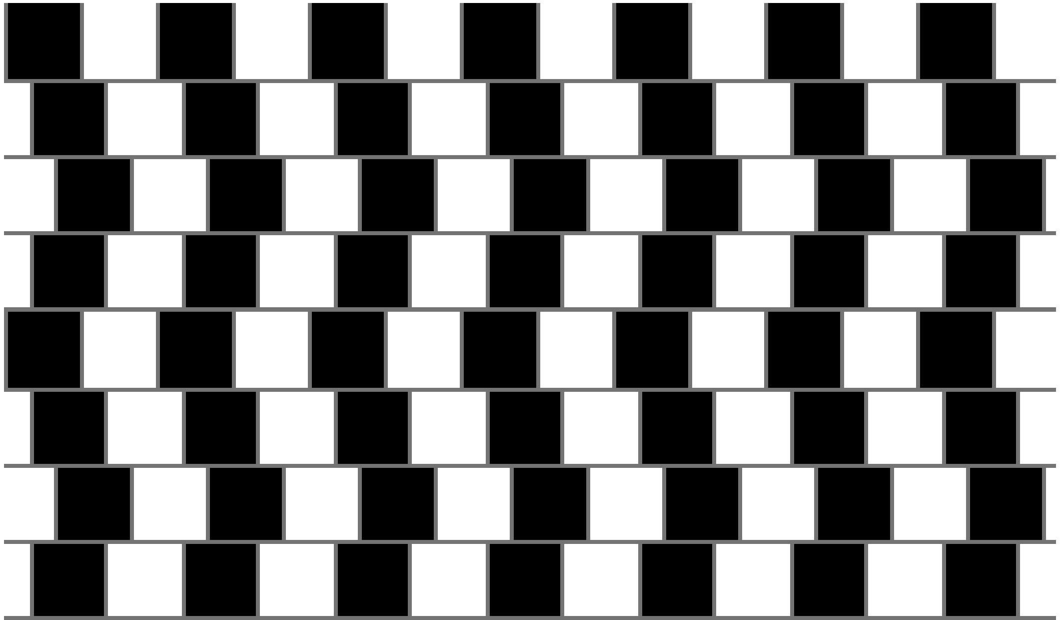
Optical Illusions (cont…)

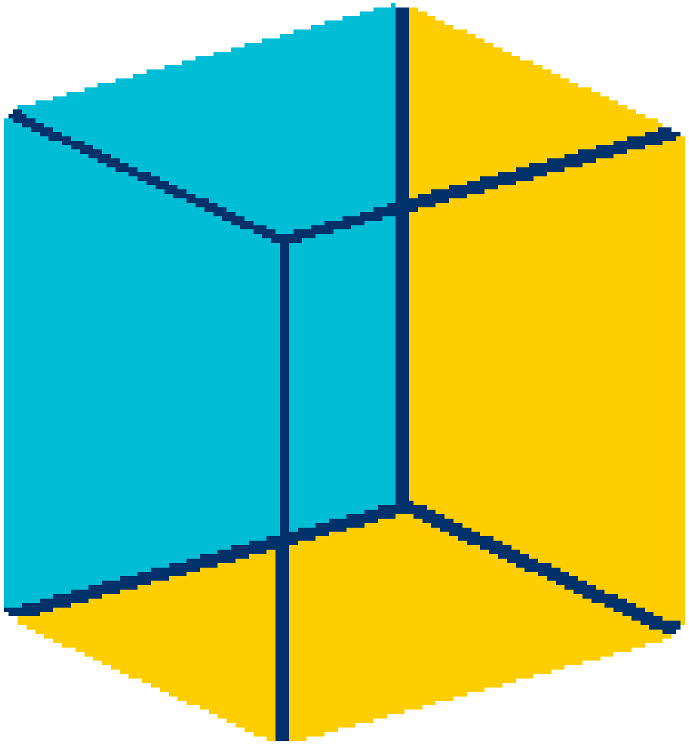


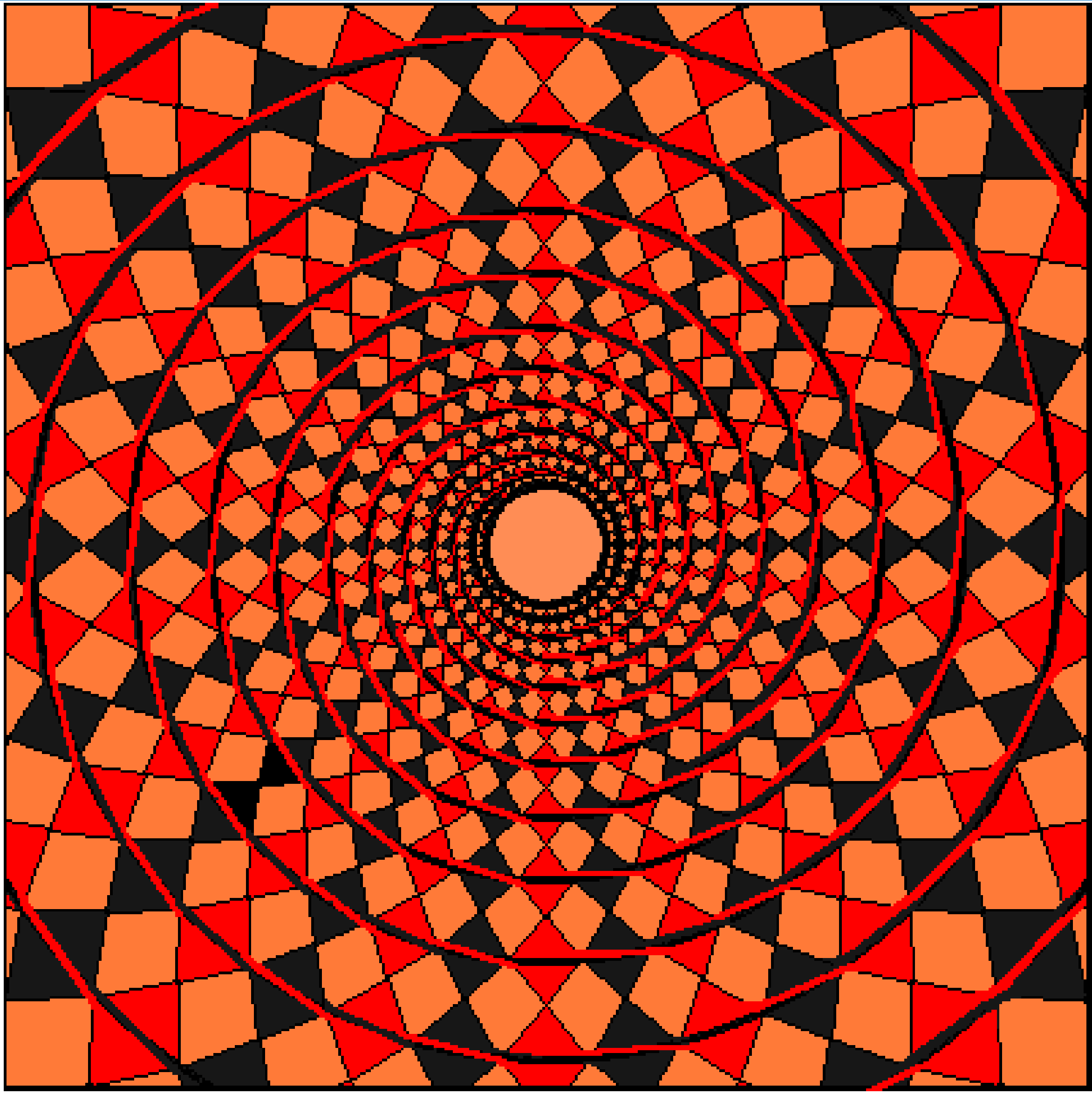
Stare at the cross in the middle of the image and think circles

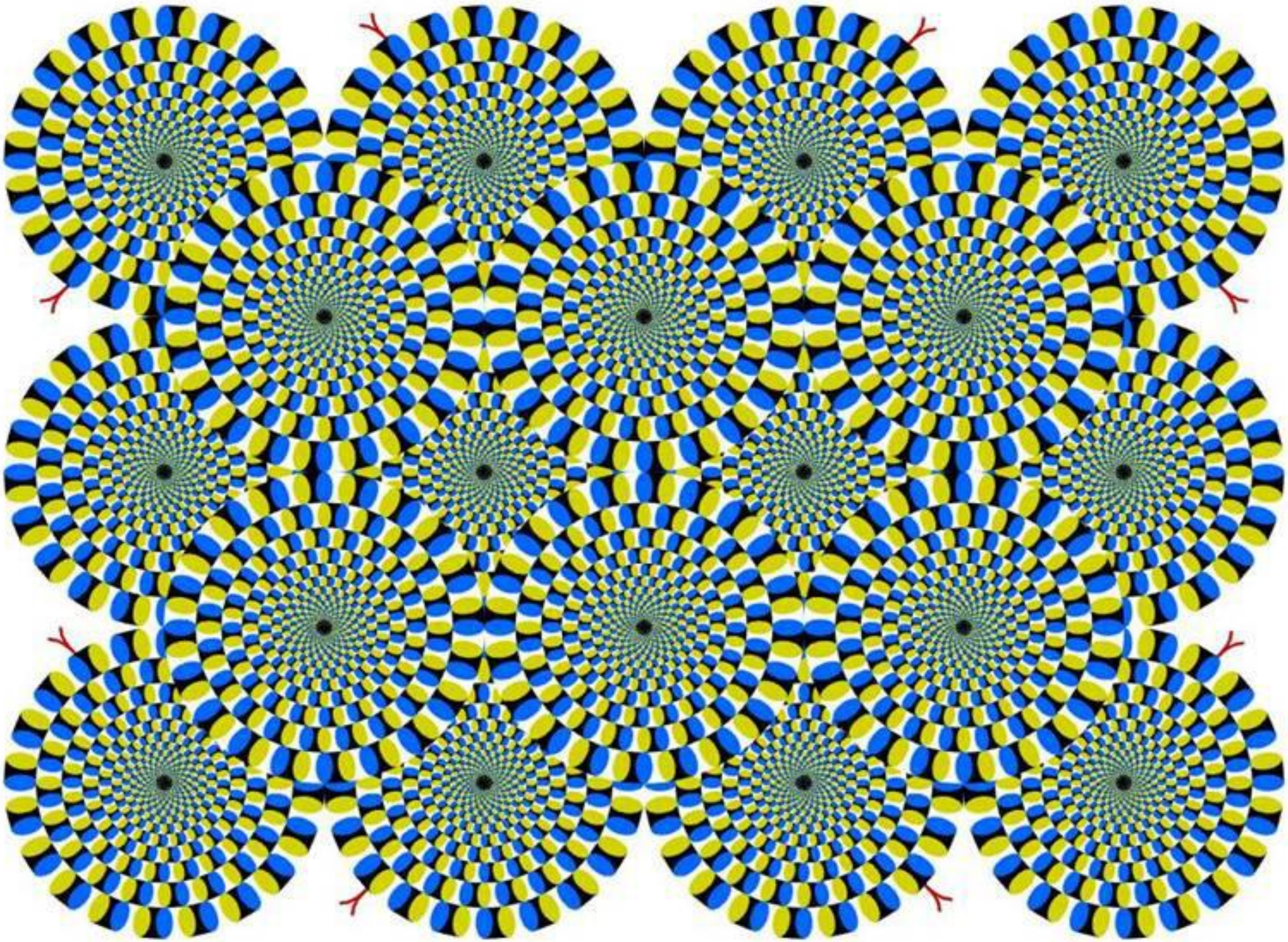


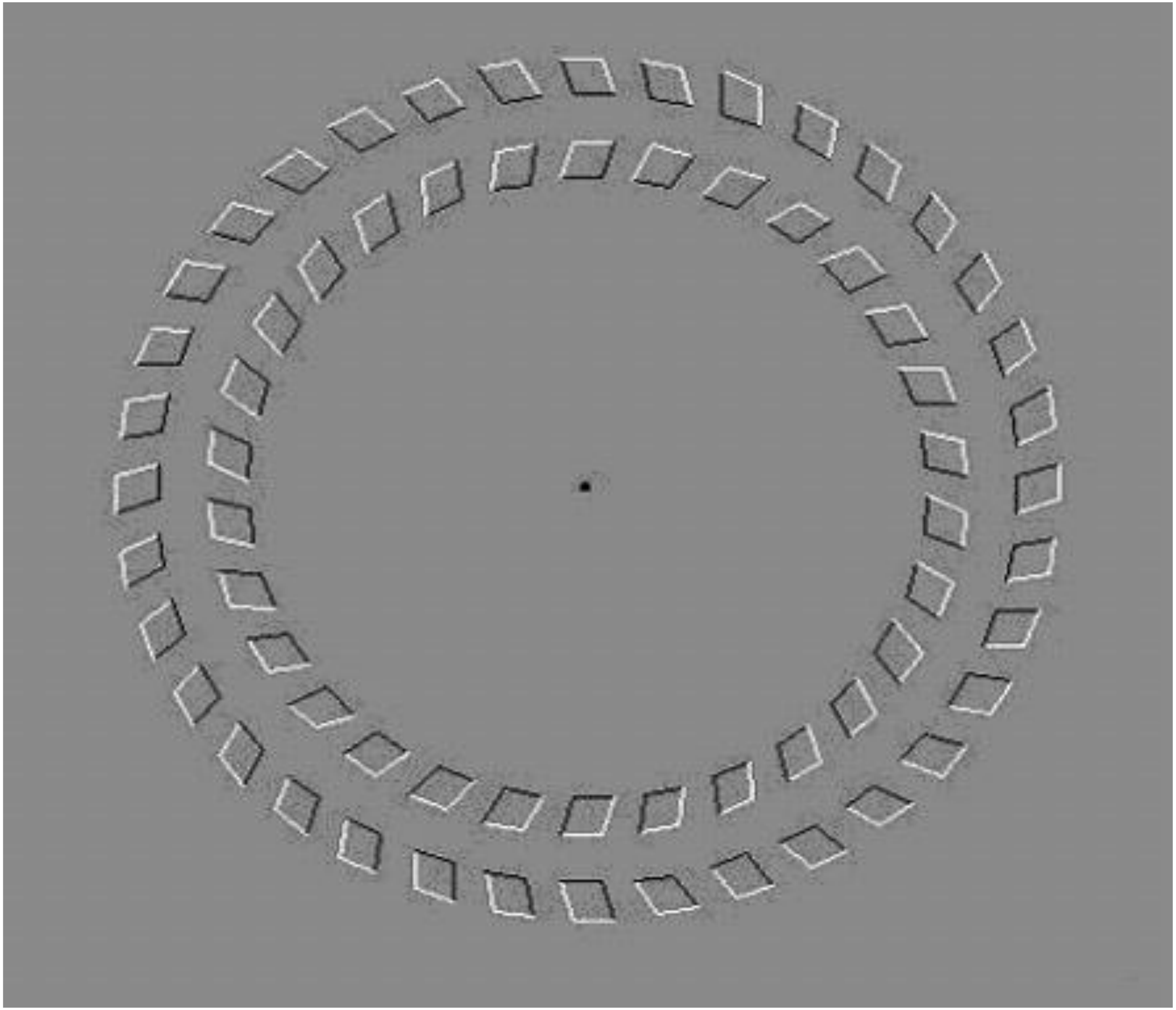


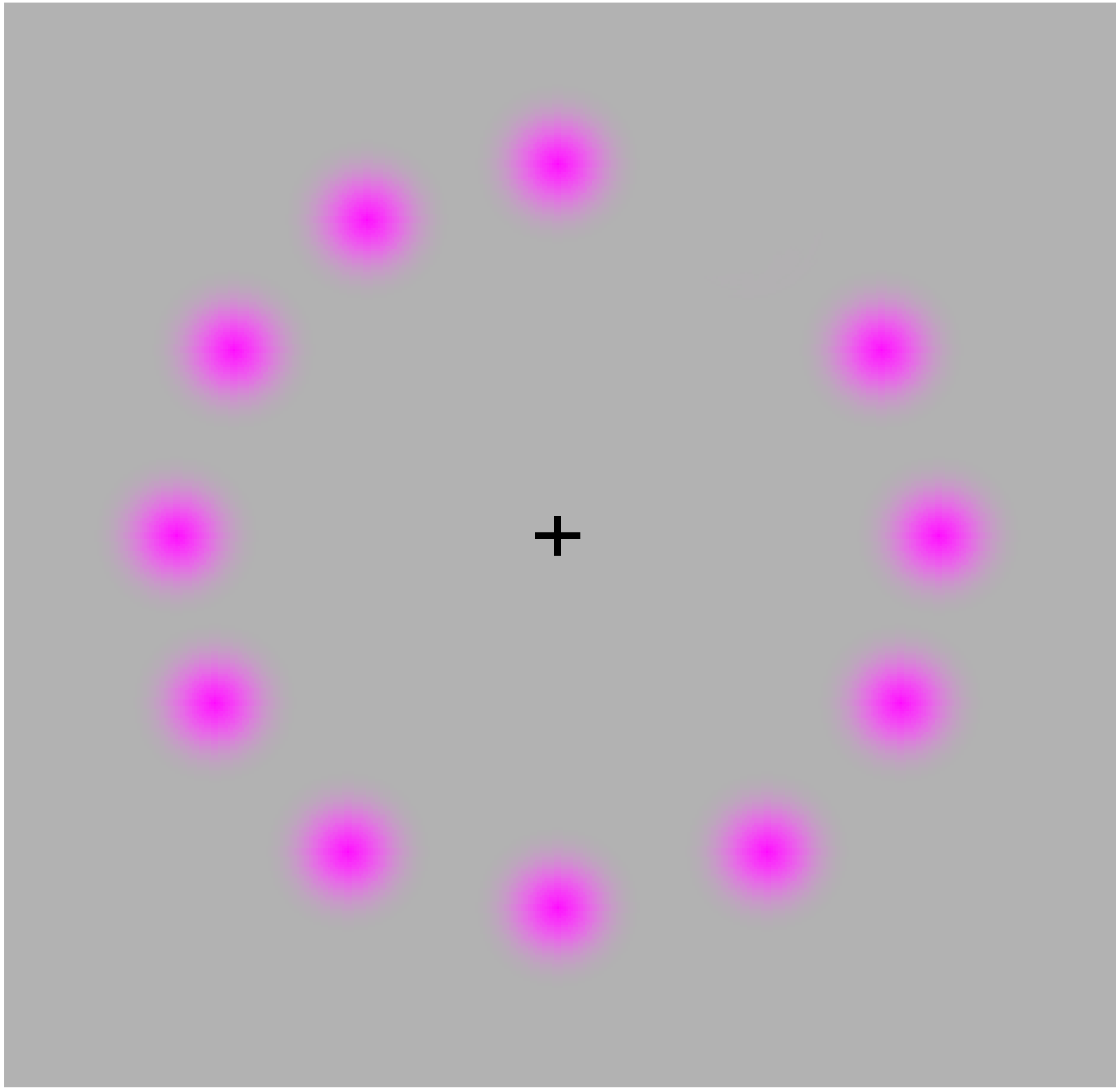












Concentrate on the cross in the middle, and the moving circle turns green! ... after a while the violet circles disappear!!

Mind Map Exercise: Mind Mapping For Note Taking

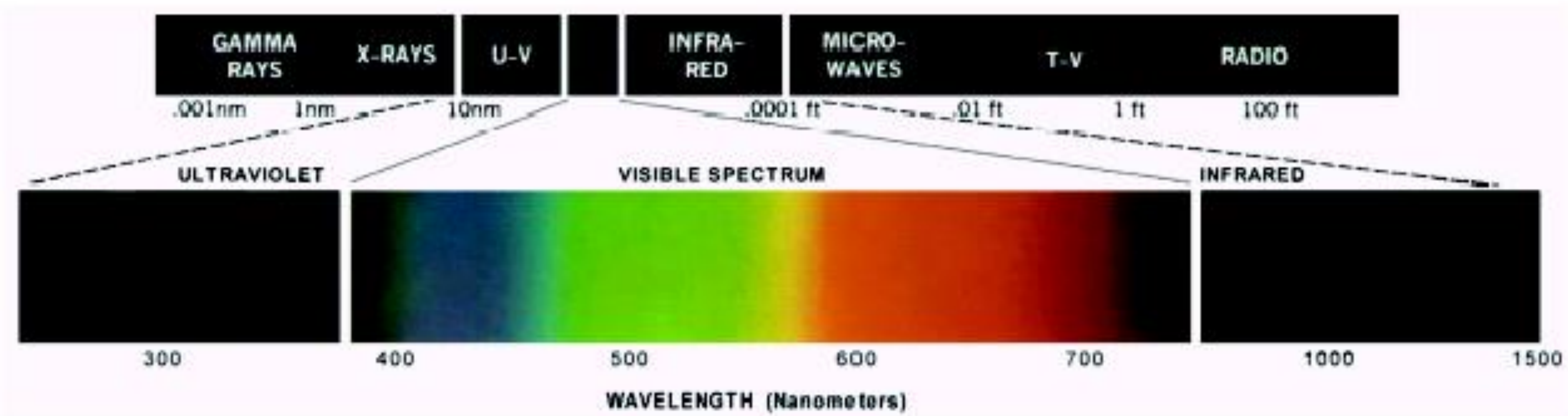
TED Ideas worth spreading

Beau Lotto: Optical Illusions Show How We See

http://www.ted.com/talks/lang/eng/beau_lotto_optical_illusions_show_how_we_see.html

2 Light and the Electromagnetic Spectrum

- Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye
- The electromagnetic spectrum is split up according to the wavelengths of different forms of energy



Light: the Visible Spectrum

achromatic or monochromatic light

- The color of an object is determined by the nature of the light *reflected* by the object
- We perceive only a small range of colors of the electromagnetic spectrum: 0.43μm(violet)-0.78μm(red)
- Six bands: violet, blue, green, yellow, orange, red
- Monochromatic light (gray level)
- Three elements measuring chromatic light
  - Radiance, luminance and brightness

Three quantities describe the quality of a chromatic light source: radiance, luminance, and brightness

Radiance(unit: W) is the total amount of energy that flows from the light source

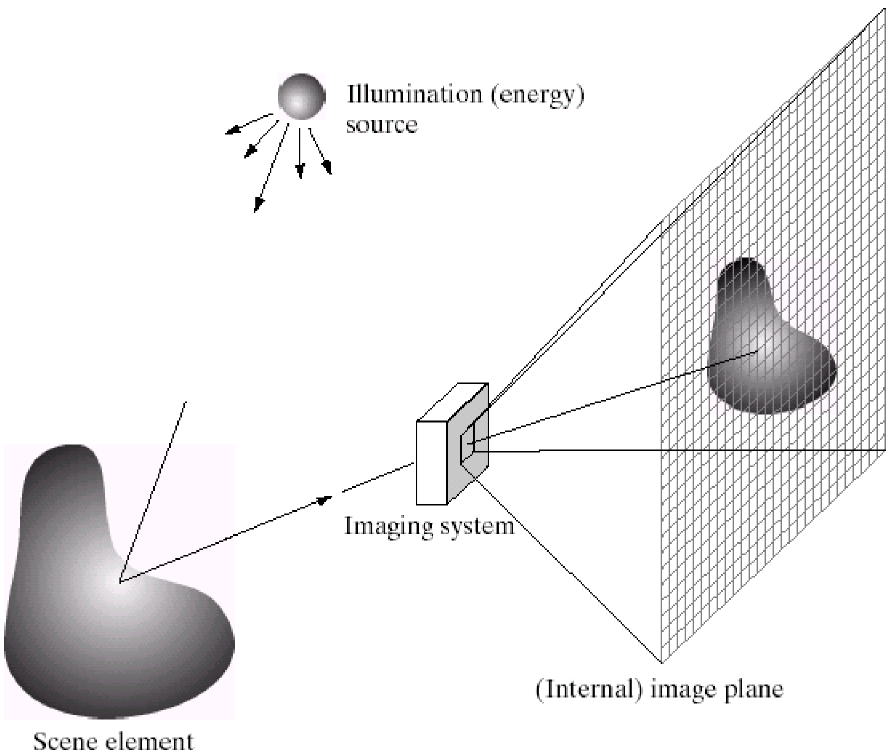
Luminance(unit: lm, lumens) gives a measure of the amount of energy an observer perceives from a light source

ex) Light emitted from a source operating in the far infrared region of the spectrum could have high radiance, but an observer would hardly perceive it(zero luminance)

Brightness is a subjective descriptor of light perception that is practically impossible to measure. It embodies the achromatic notion of intensity

3 Image sensing and acquisition

Images are typically generated by *illuminating* a *scene* and absorbing the energy reflected by the objects in that scene



Typical notions of illumination and scene can be way off:

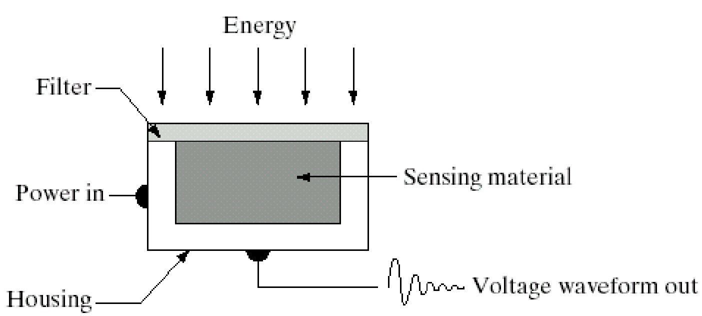
a) X-rays of a skeleton
b) Ultrasound of an unborn baby
c) Electro-microscopic images of molecules

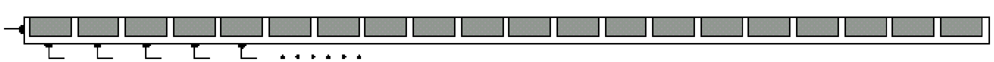
Gonzalez & Woods, Digital Image Processing (2010)

1) Image Sensing

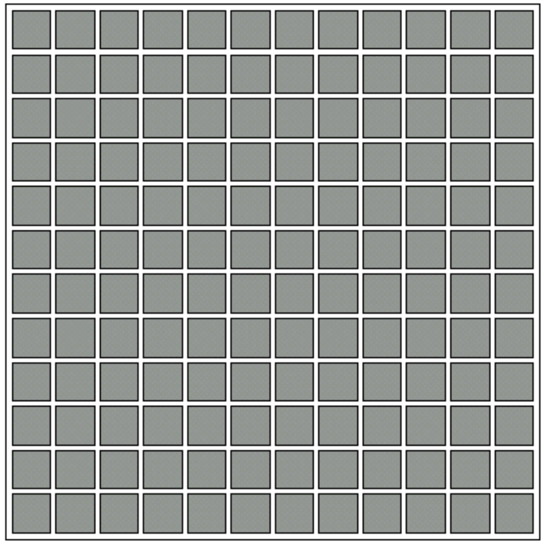
Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected. Three sensor arrangements:

Imaging Sensor



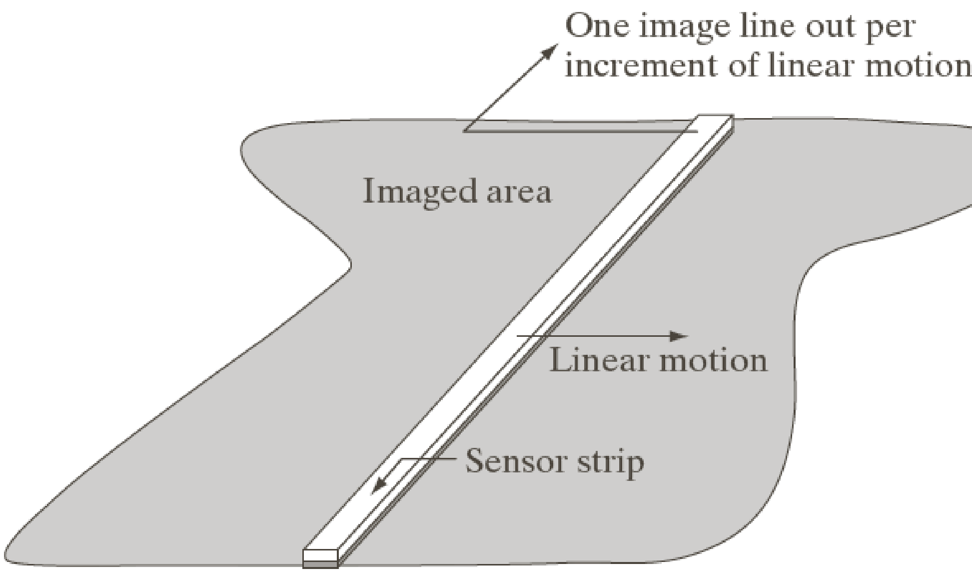


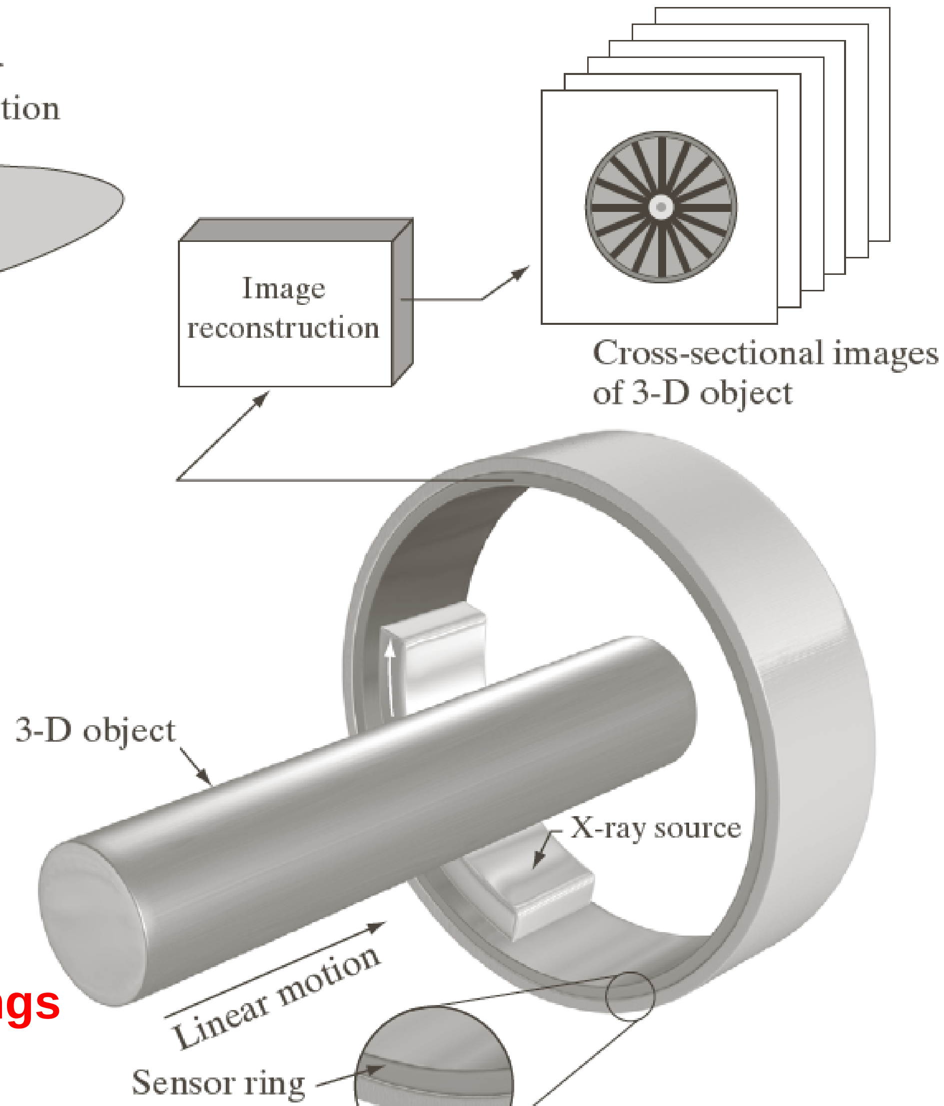
Line of Image Sensors



Array of Image Sensors

Image Sensing

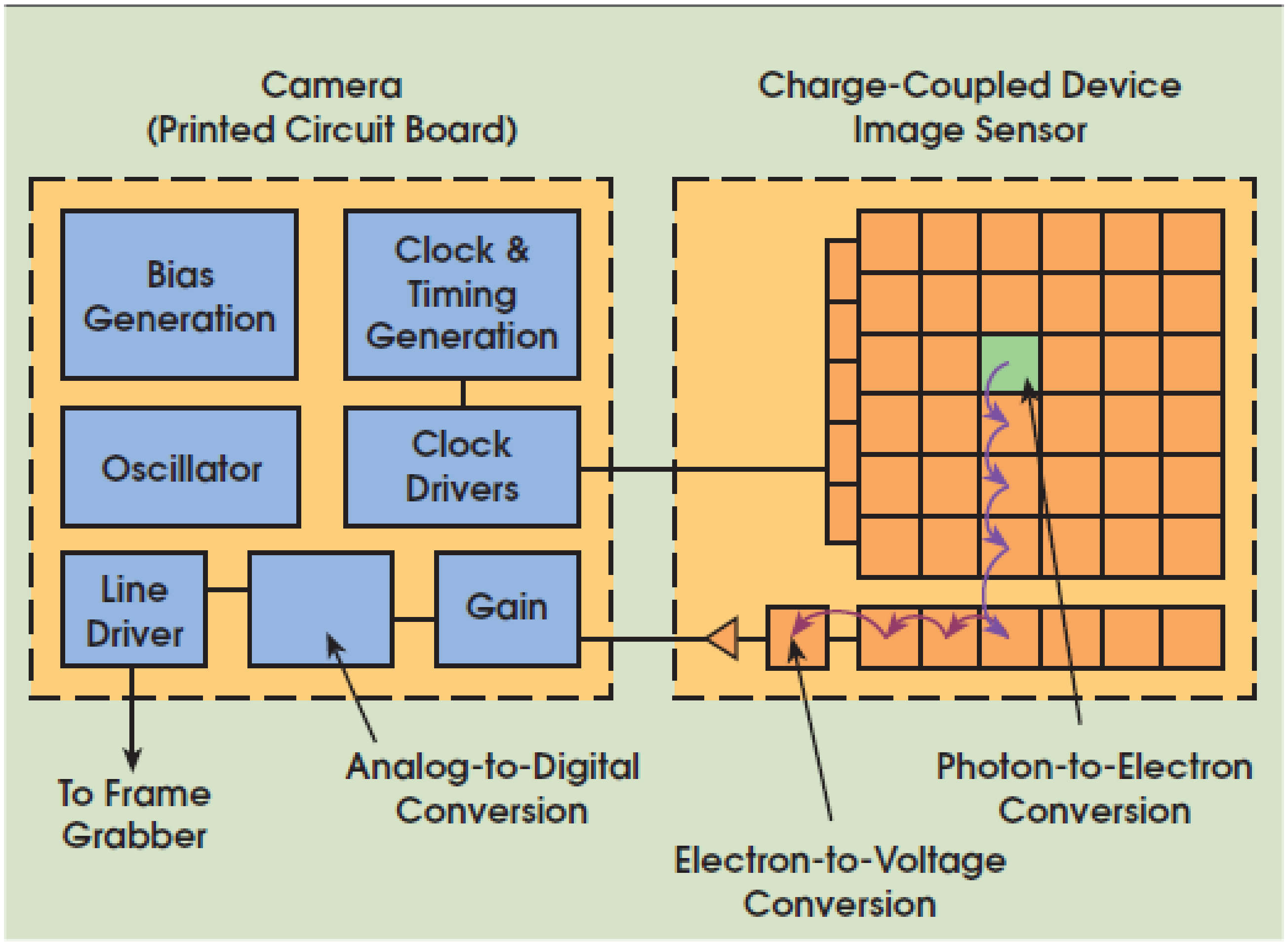




Using Sensor Strips and Rings

Gonzalez & Woods, Digital Image Processing (2010)

Charge coupled device (CCD) image sensor



The Nobel Prize in Physics 2009

"for groundbreaking achievements concerning the transmission of light in fibers for optical communication"

"for the invention of an imaging semiconductor circuit – the CCD sensor"

Photo: U. Montan

Charles K. Kao

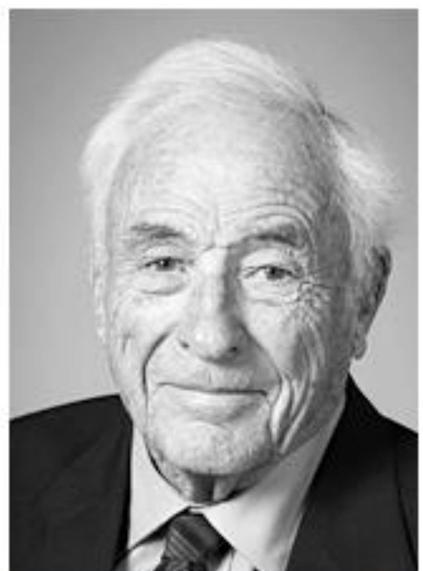


Photo: U. Montan

Willard S. Boyle

Photo: U. Montan

George E. Smith

Complementary Metal Oxide Semiconductor (CMOS) Sensor

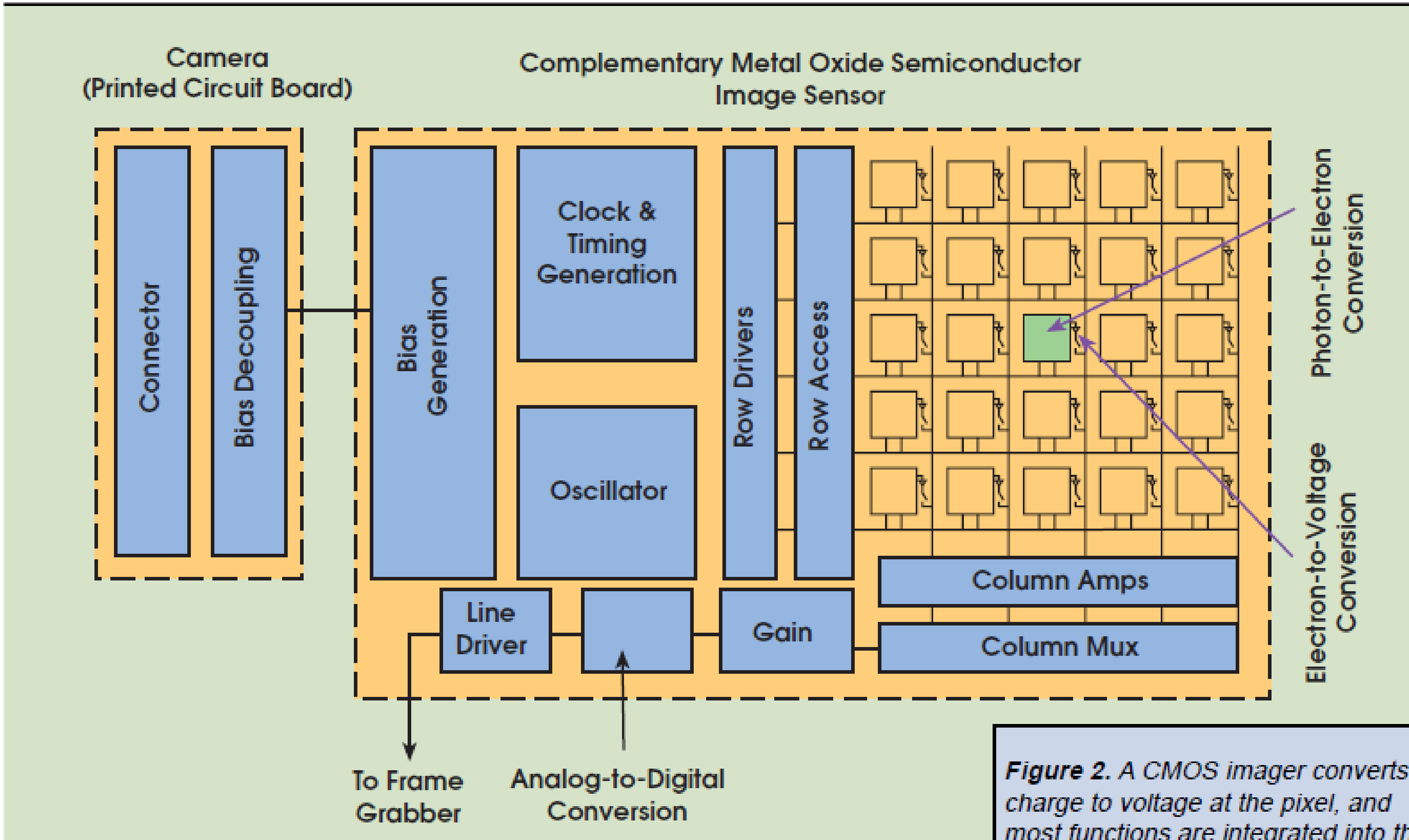
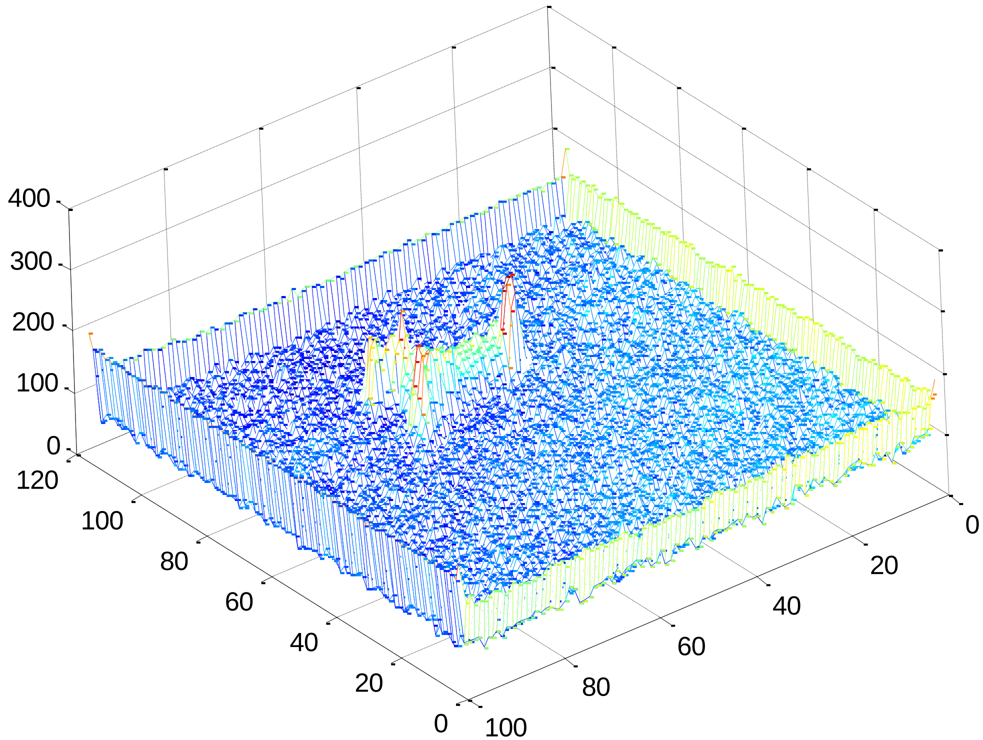
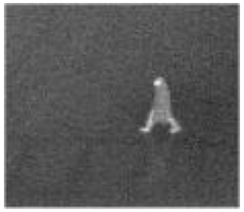


Figure 2. *A CMOS imager converts charge to voltage at the pixel, and most functions are integrated into the*

3D Visualization





It is useful to take an analogy to rain gauge (image intensity values
Measure the amount of ``photon rain”)

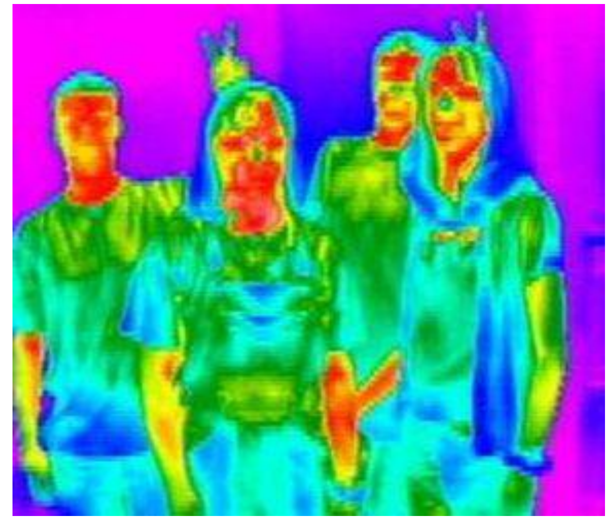
High Dynamic Range Imaging

HDR Display (After Toner Mapping)

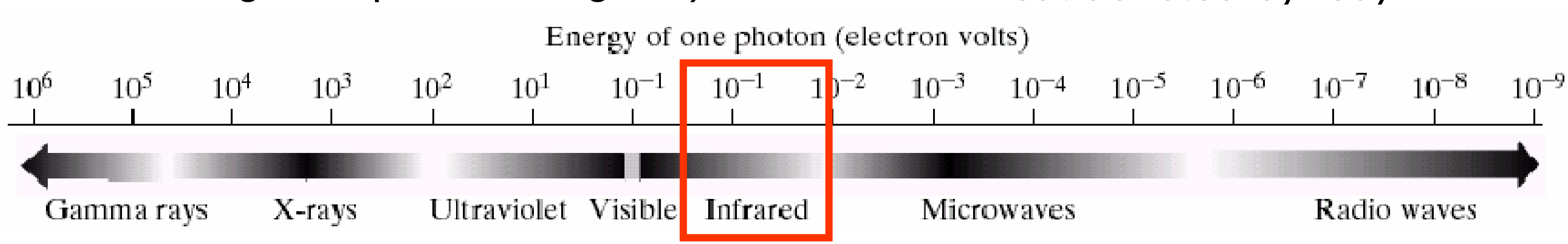
Thermal Imaging

Operate in infrared frequency

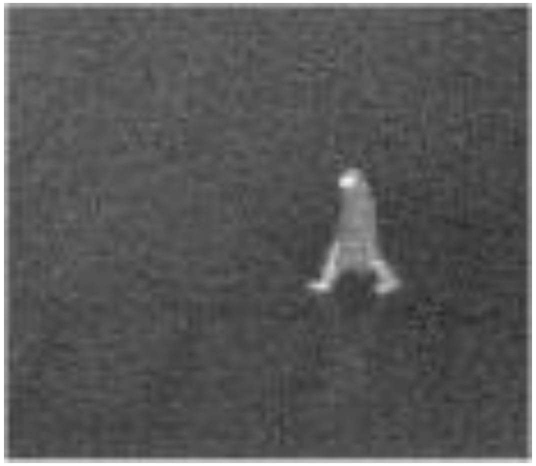
Grayscale representation
(bright pixels correlate with
high-temperature regions)

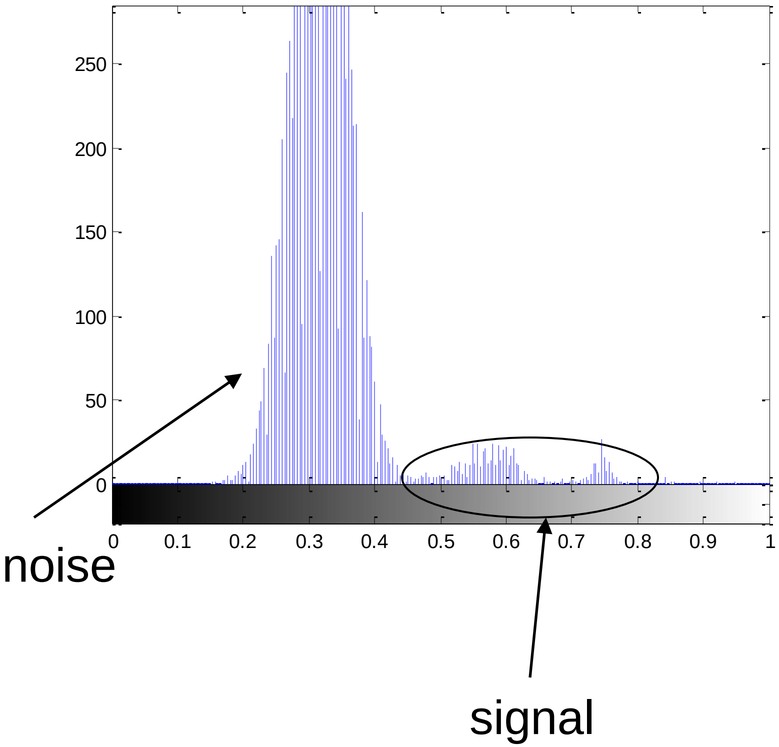


Pseudo-color representation
(Human body dispersing
heat denoted by red)



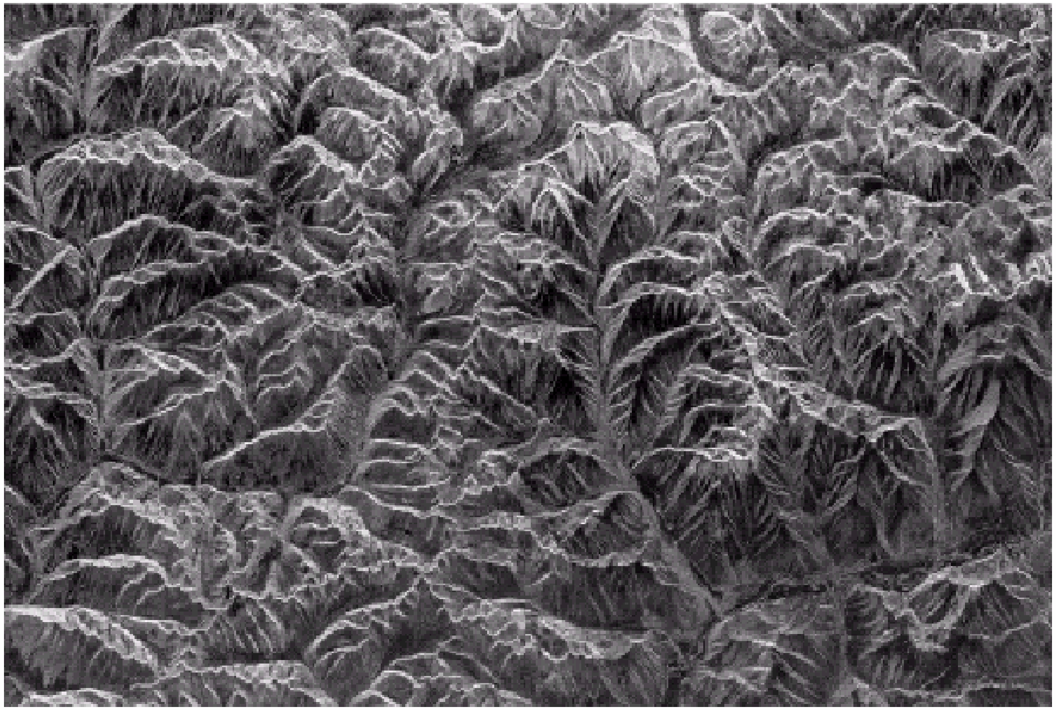
Low Signal-to-Noise (SNR) Behavior

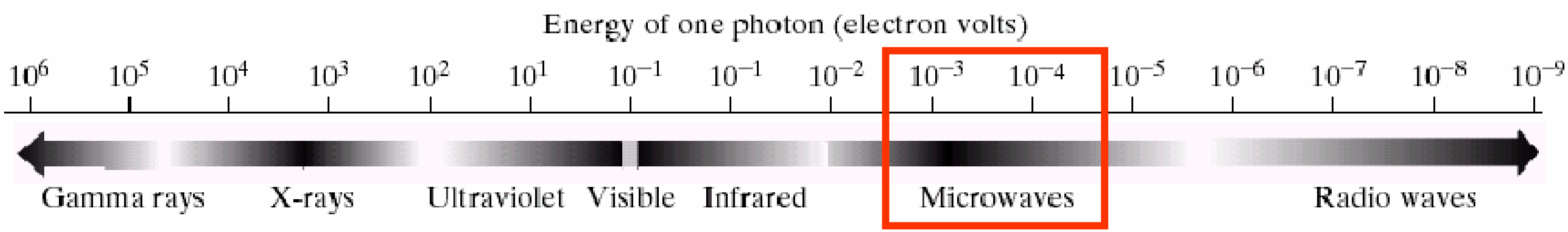




Radar Imaging

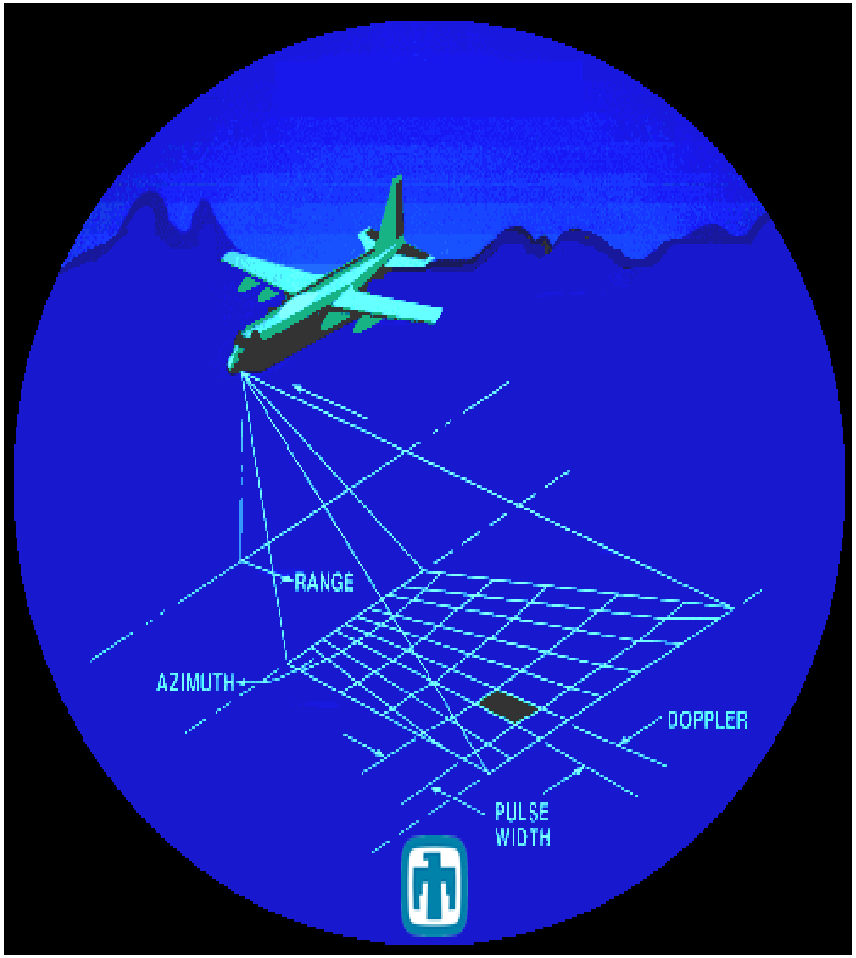
Operate in microwave frequency





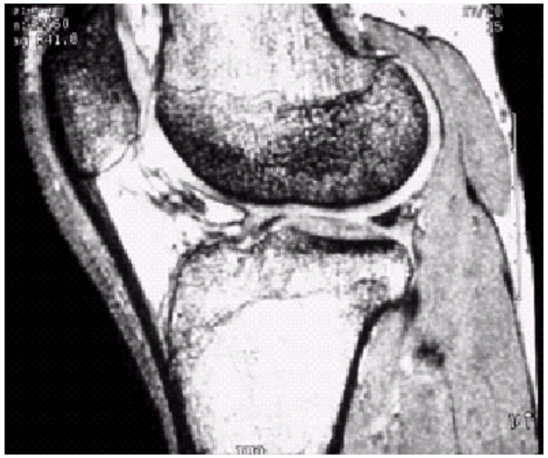
Synthetic Aperture Radar (SAR)

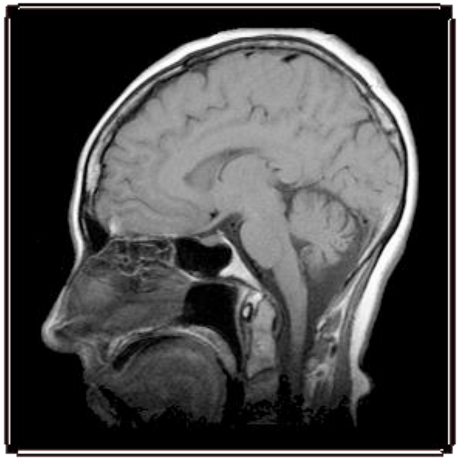
- Environmental monitoring, earth-resource mapping, and military systems
- SAR imagery must be acquired in inclement weather and all-day-all-night.
- SAR produces relatively fine resolution that differentiates it from other radars.

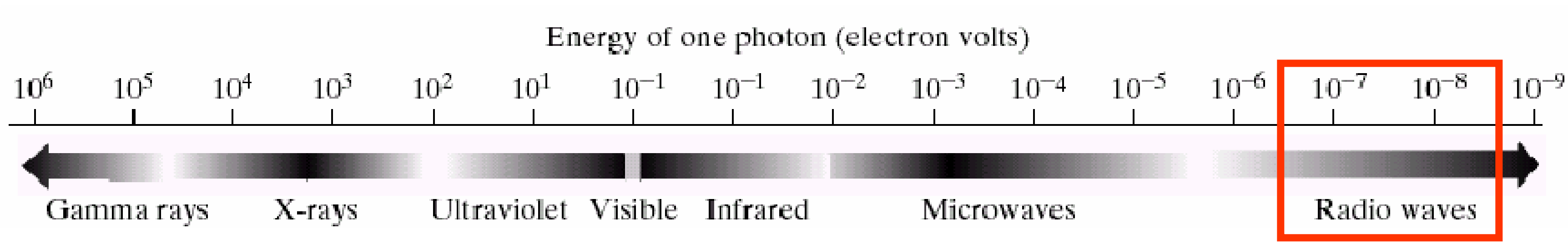


Magnetic Resonance Imaging (MRI)

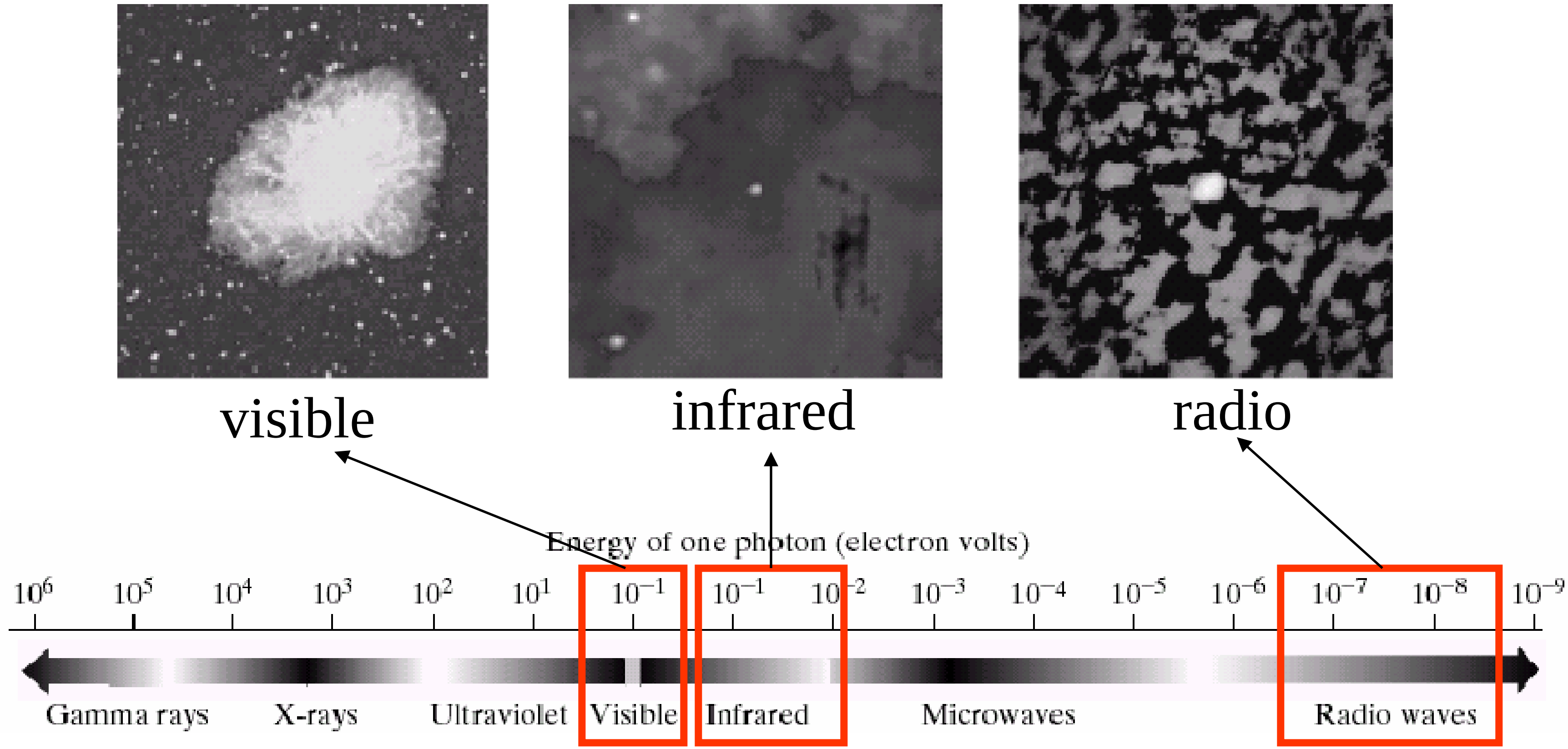
Operate in radio frequency





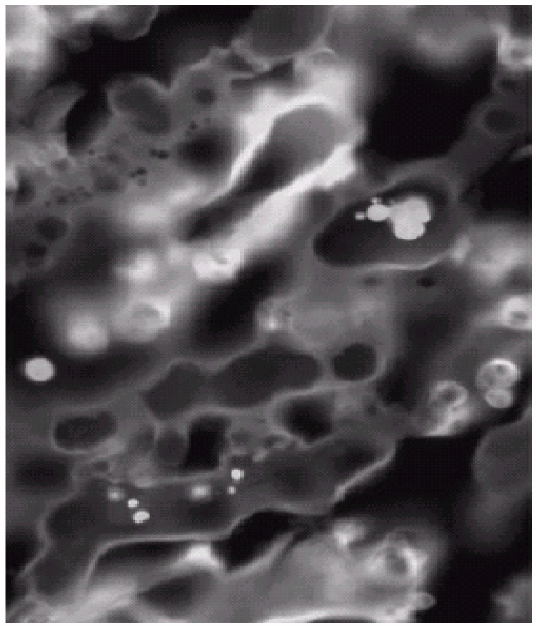


Comparison of Different Imaging Modalities

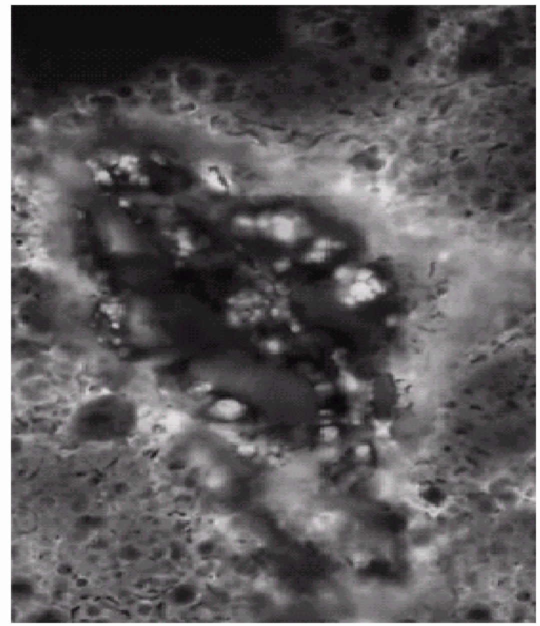


Fluorescence Microscopy Imaging

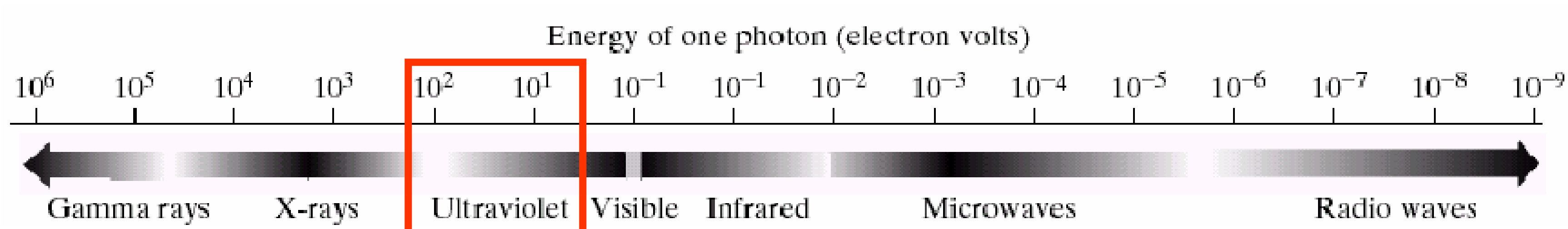
Operate in ultraviolet frequency



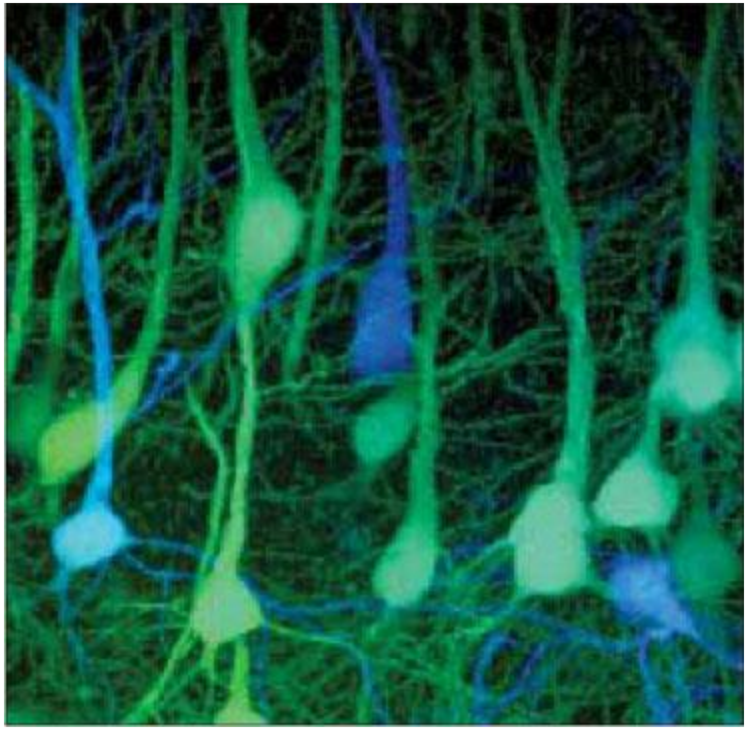
normal corn



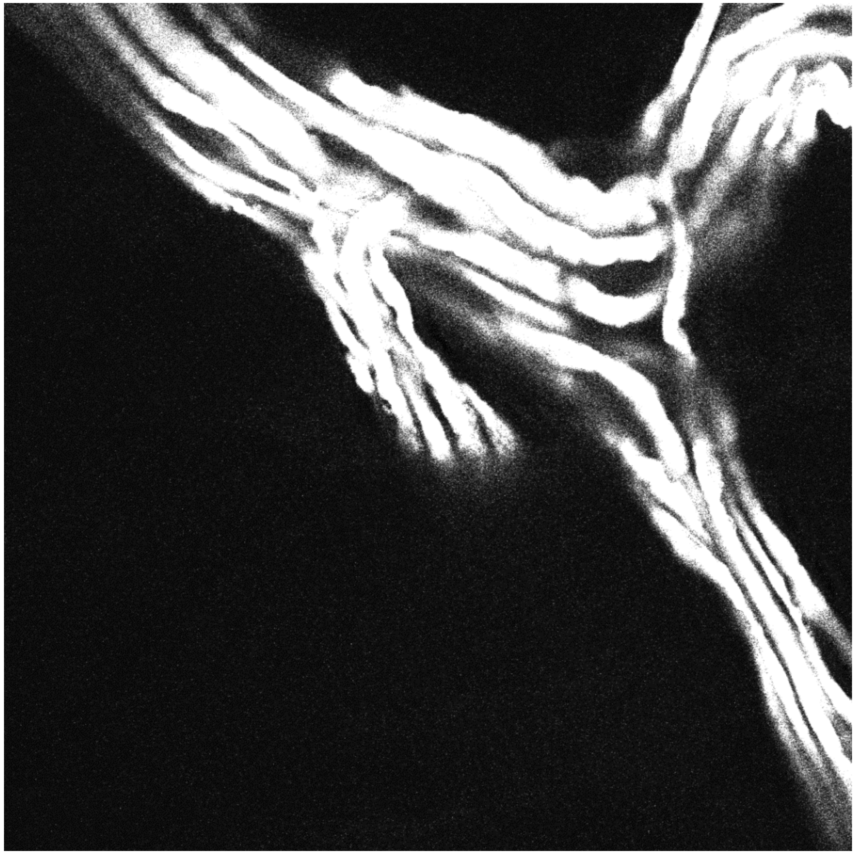
smut corn



What Does a Neuron Look Like?

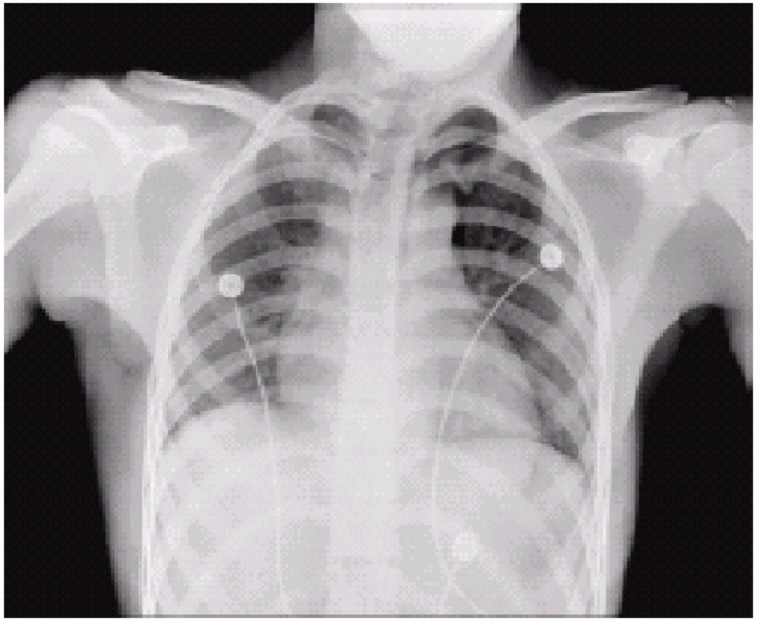


Artistic illustration



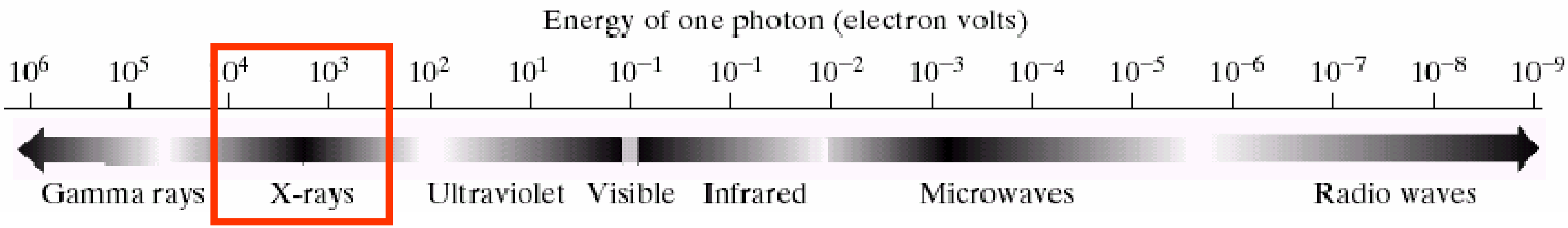
Real image

X-ray Imaging

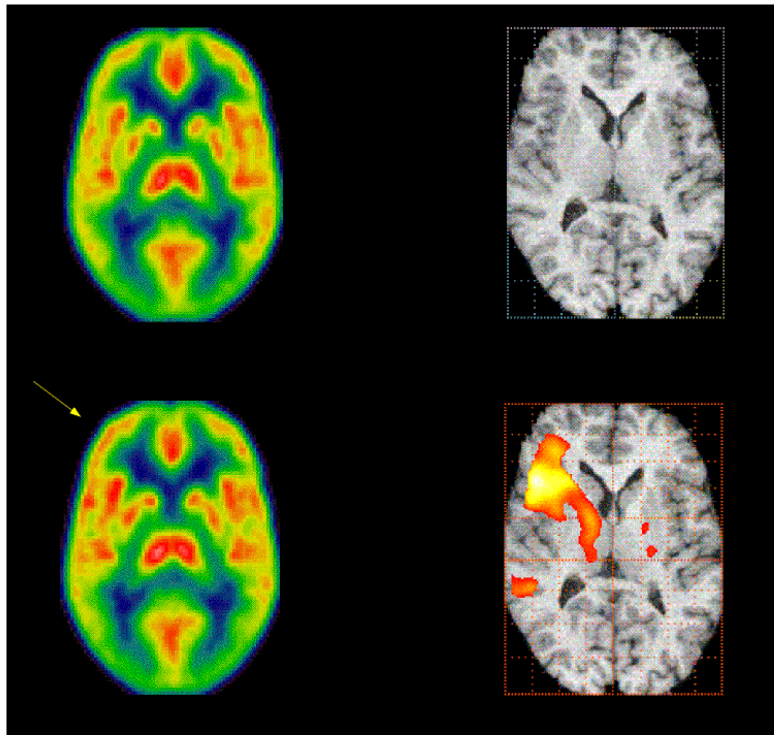


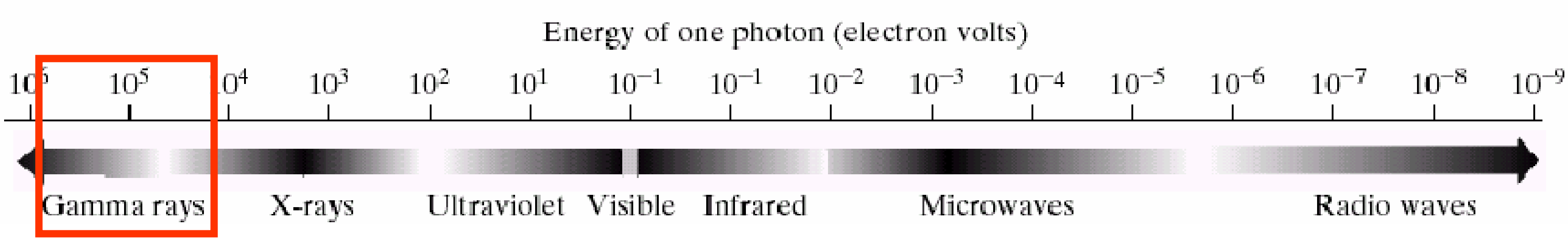
chest

head



Operate in gamma-ray frequency





2) Simple Image Formation Model

Monochrome(Gray) Image : $f(x,y)$
$f(x,y)$ = ***intensity value*** at coordinates (x,y)

$0 < f(x,y) < \infty$; $f(x,y)$ is energy

Simple image formation

$$f(x,y) = i(x,y)r(x,y)$$

$$0 < i(x,y) < \infty \text{ ; illumination}$$

$$0 < r(x,y) < 1 \text{ ; reflectance}$$

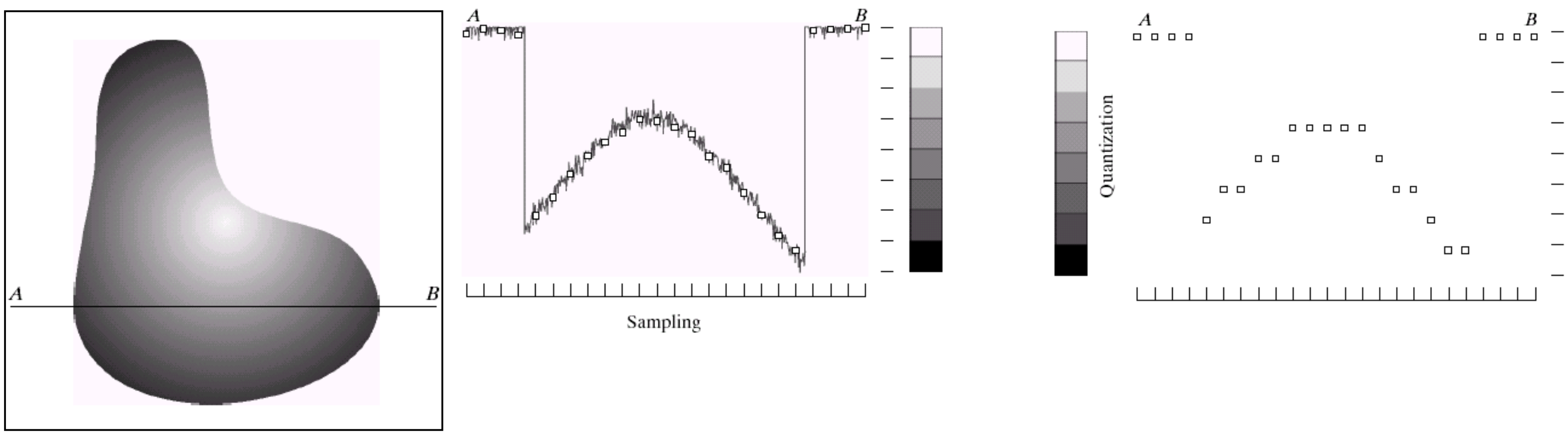
In real situation (level)

$$L_{min} \le l\ (=f(x,y)) \le L_{max}$$

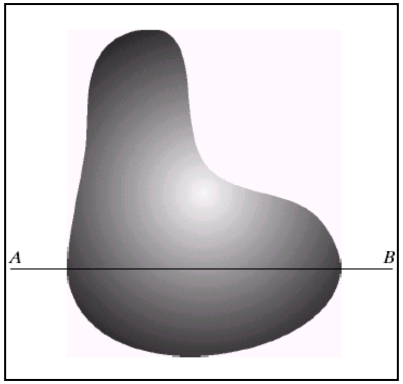
4 Image sampling, quantization and representation

A digital sensor can only measure a limited number of **samples** at a **discrete** set of energy levels

Quantization is the process of converting a continuous **analogue** signal into a digital representation of this signal



4.1 Image Sampling and Quantisation



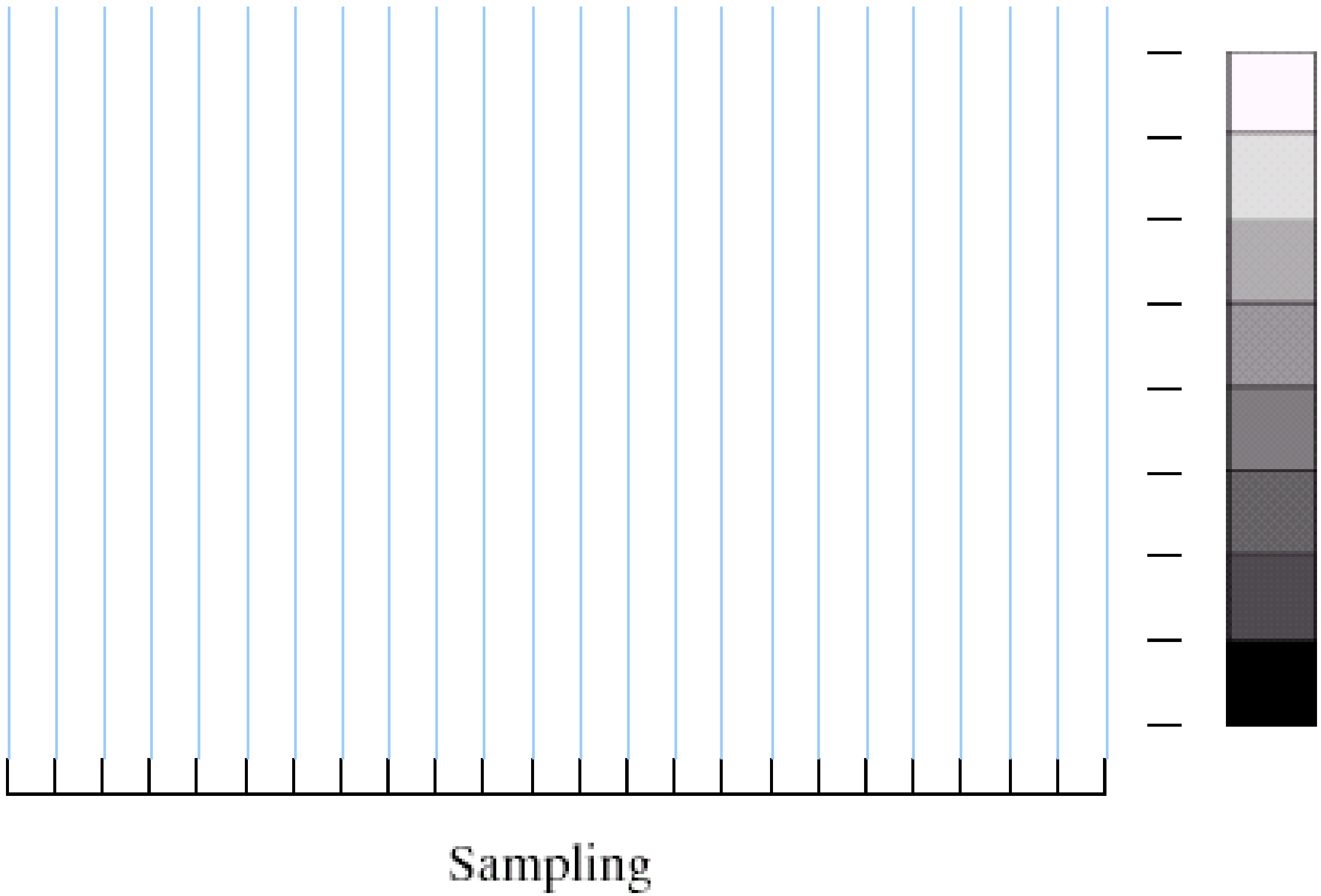
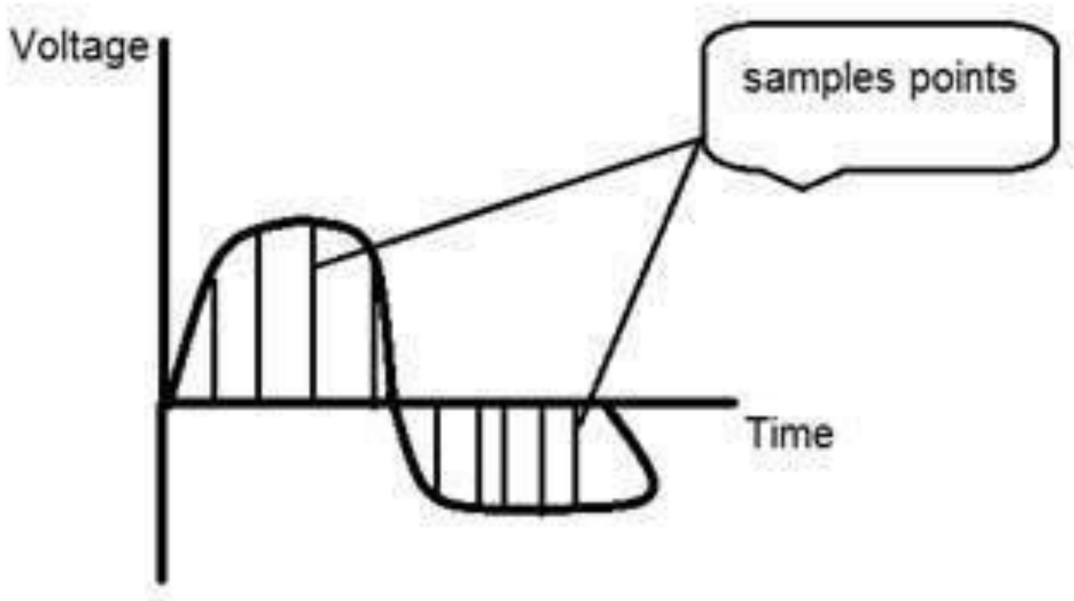
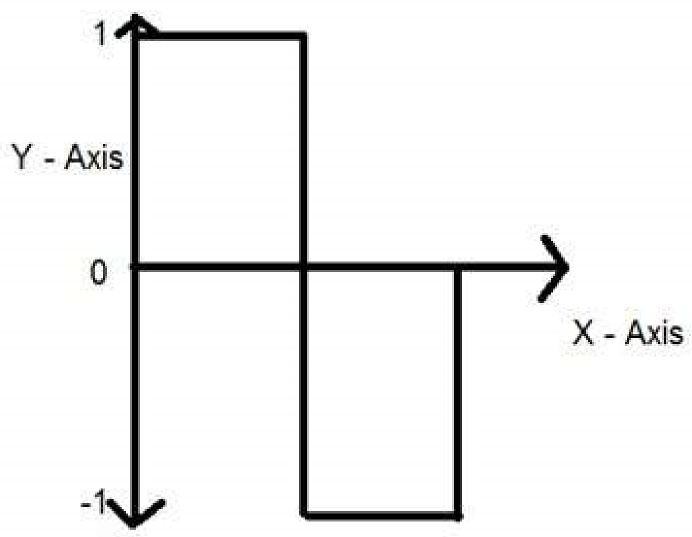


Image Sampling



Sampling is done on x axis.
It is the conversion of x axis (infinite values) to digital values.

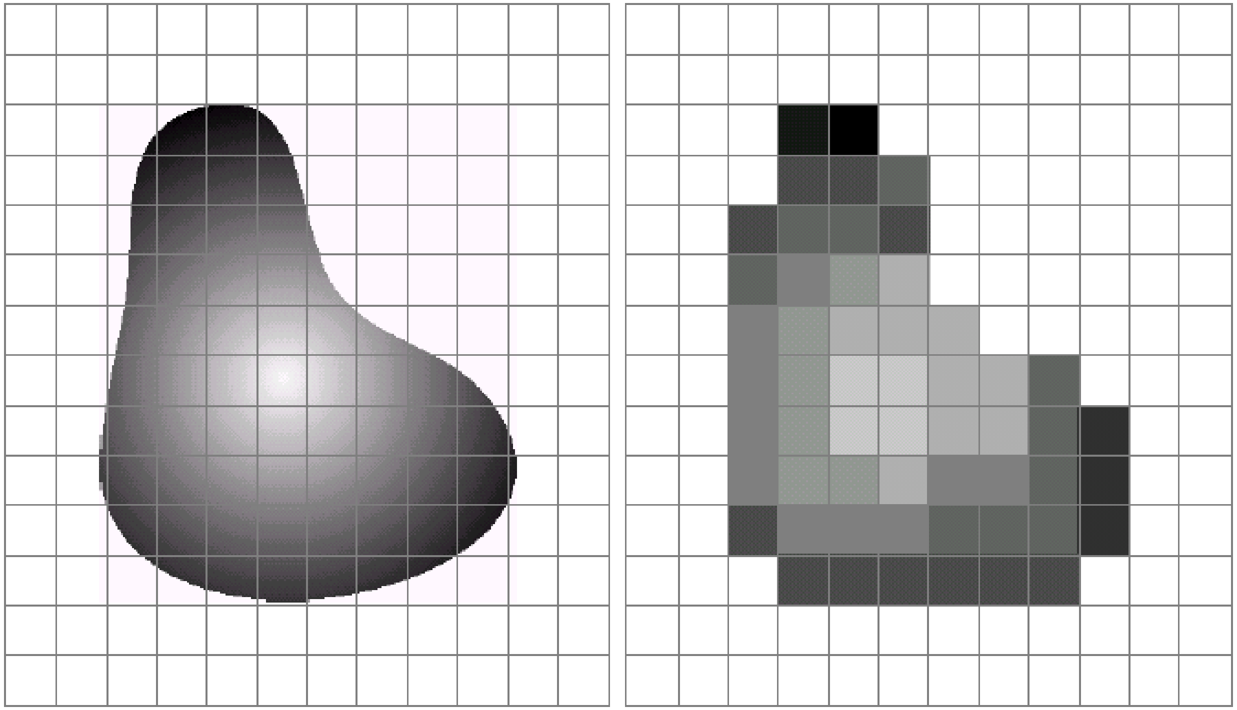
Quantization



It is done on y axis. When you are quantizing an image, you are actually dividing a signal into quanta(partitions). On the x axis of the signal, are the co-ordinate values, and on the y axis, we have amplitudes.
So digitizing the amplitudes is known as Quantization.

4.1 Image Sampling and Quantisation

Remember that a digital image is always only an **approximation** of a real world scene



4.2 Image representation

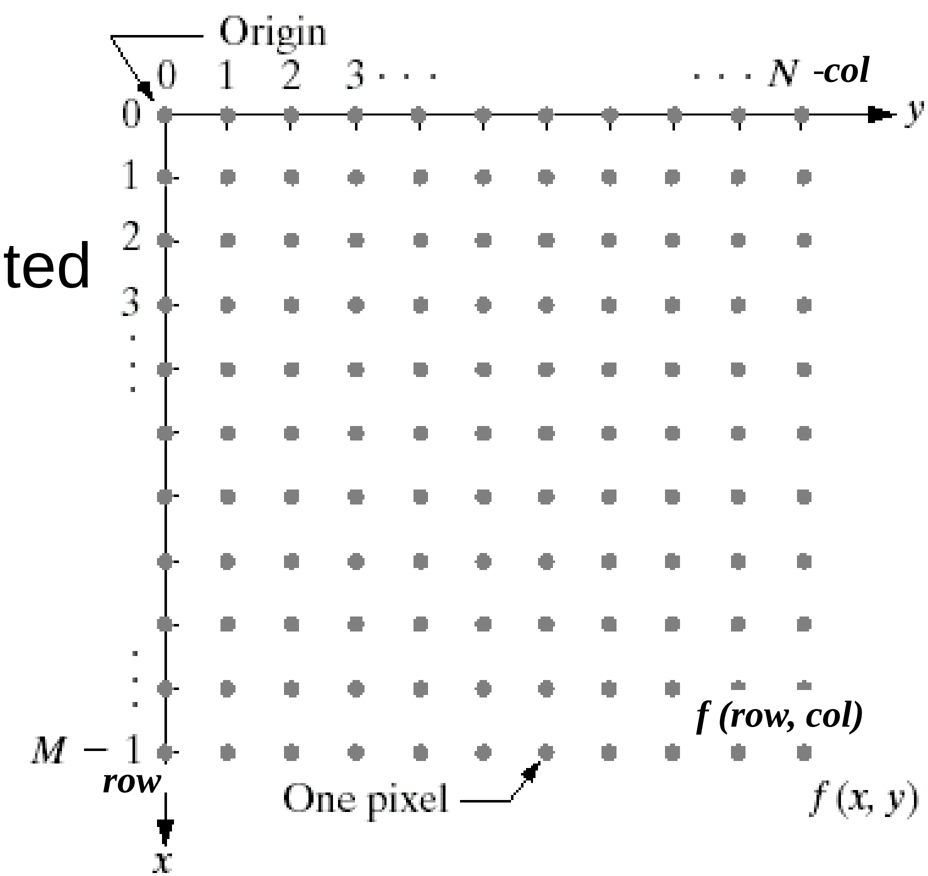
- A digital image is composed of M rows and N columns of pixels each storing a value.
- Pixel values are most often grey levels in the range 0-255 (black-white).
- Images can easily be represented as matrices.

Three ways to represent:

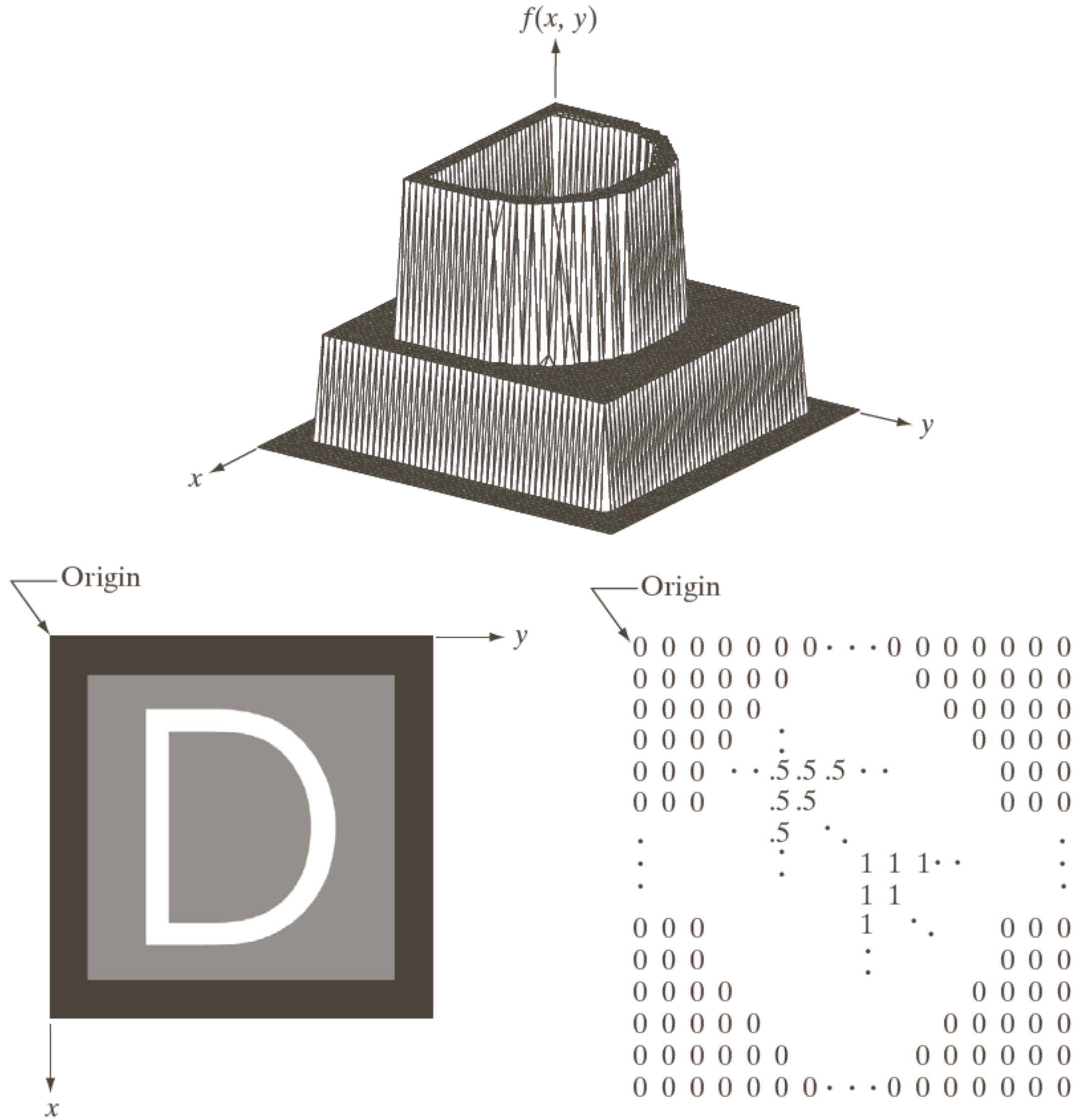
(1) Plotted as a surface;

(2) A visual intensity array;

(3) 2-D numerical array



4.2 Image representation



5 Image Resolution

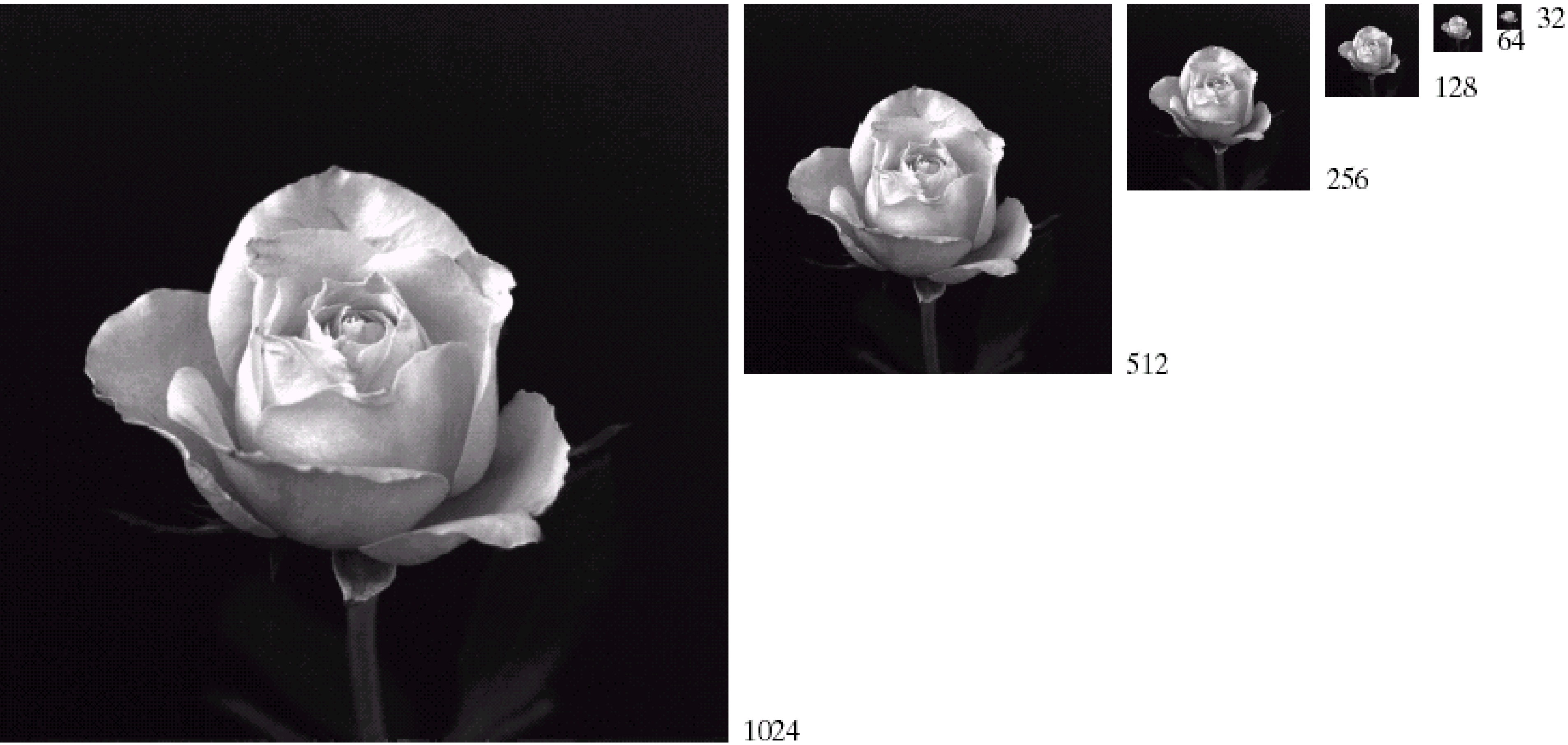
5.1 Spatial Resolution

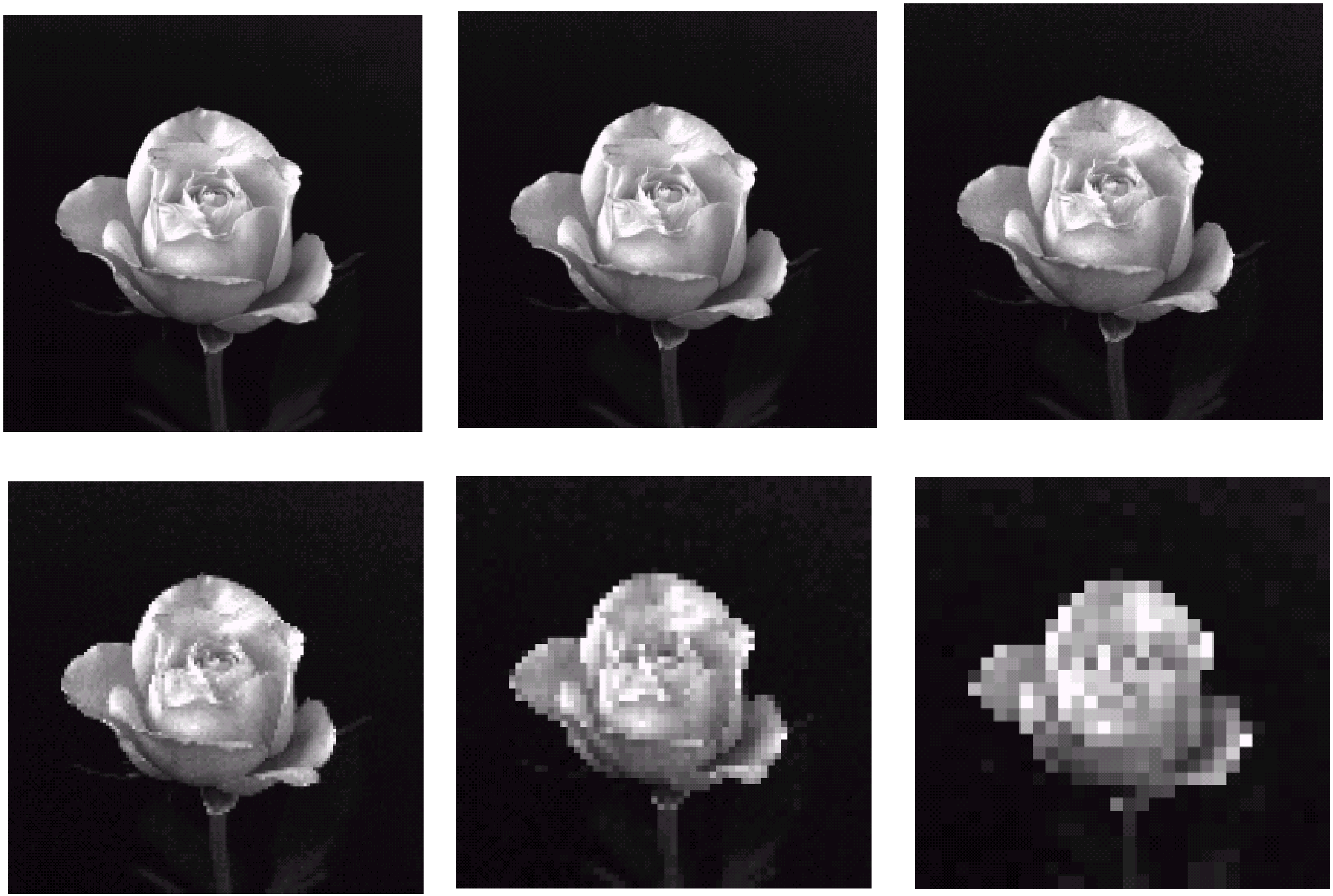
Spatial resolution of an image is determined by how sampling was carried out.

Spatial resolution simply refers to the smallest discernable detail in an image.

- Vision specialists often talk about pixel size
- Graphic designers use *Dots Per Inch* (DPI)[US]
- *newspaper 75 DPI, magazines 133 DPI, book page 2400 DPI*

Spatial Resolution



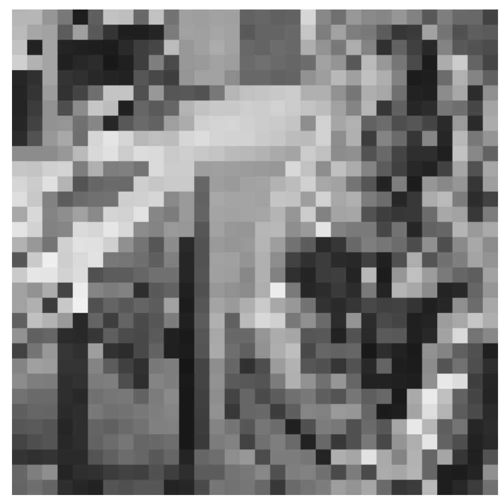


original

Sampled by 2

Sampled by 4

Sampled by 8



Sampled by 16

5.2 Intensity Level Resolution

Intensity level resolution refers to the number of intensity levels used to represent the image

- The more intensity levels used, the finer the level of detail discernable in an image
- Based on the hardware considerations, the number of intensity levels is an integer power of 2
- Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

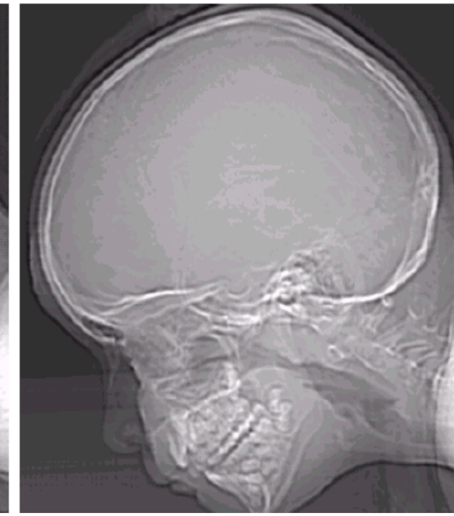
Intensity Level Resolution

256 grey levels (8 bits per pixel)

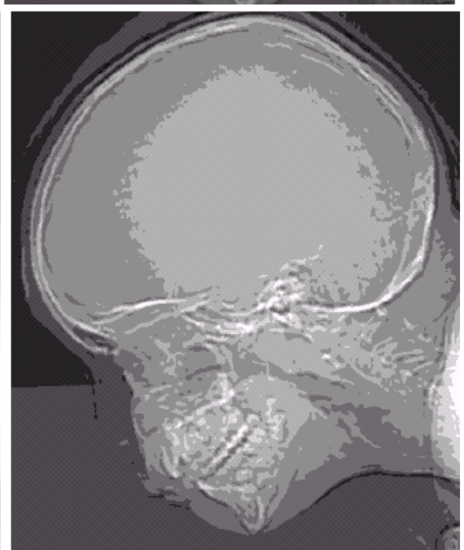
128 grey levels (7 bpp)

64 grey levels (6 bpp)

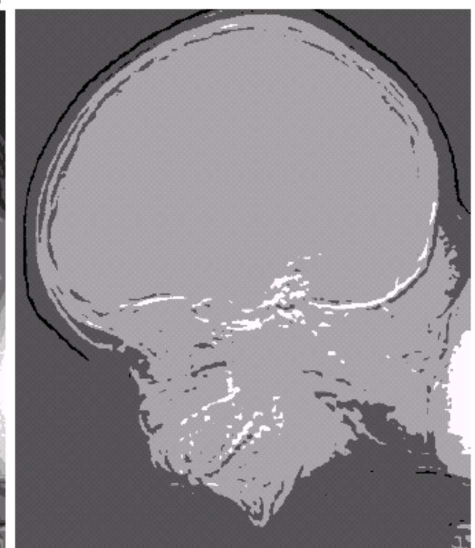
32 grey levels (5 bpp)



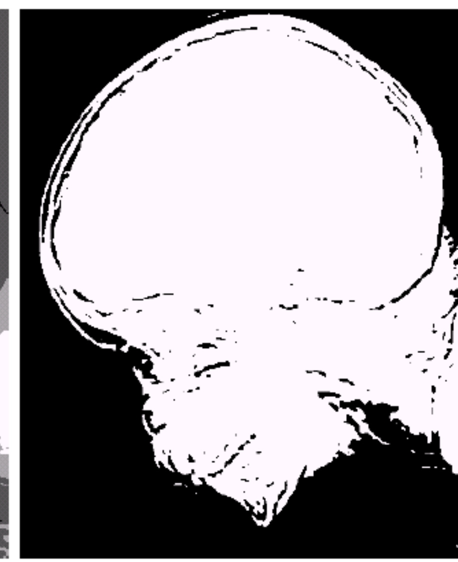
16 grey levels (4 bpp)



8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)

original

128 levels (7 bits)

16 levels (4 bits)

4 levels (2 bits)

2 levels (1 bit)

Resolution: How Much Is Enough?

– This all depends on what is in the image and what you would like to do with it

– Key questions include

  - Does the image look pleasing?
  - Can you see what you need to see within the image?

counting the number of cars?

reading the number plate?

Intensity Level Resolution

Images with a large amount of detail only a few intensity levels may be needed.

Low Detail

Medium Detail

High Detail

Summary

We have looked at:

- Human visual system
- Light and the electromagnetic spectrum
- Image sensing and acquisition
- Image representation
- Sampling, quantisation and resolution

Next time we start to look at techniques for image enhancement